

Secretion of the fungal toxin candidalysin is dependent on conserved precursor peptide sequences

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The opportunistic fungal pathogen *Candida albicans* damages host cells via its peptide toxin, candidalysin. Before secretion, candidalysin is embedded in a precursor protein, Ece1, which consists of a signal peptide, the precursor of candidalysin and seven non-candidalysin Ece1 peptides (NCEPs), and is found to be conserved in clinical isolates. Here we show that the Ece1 polypeptide does not resemble the usual precursor structure of peptide toxins. *C. albicans* cells are not susceptible to their own toxin, and single NCEPs adjacent to candidalysin are sufficient to prevent host cell toxicity. Using a series of Ece1 mutants, mass spectrometry and anti-candidalysin nanobodies, we show that NCEPs play a role in intracellular Ece1 folding and candidalysin secretion. Removal of single NCEPs or modifications of peptide sequences cause an unfolded protein response (UPR), which in turn inhibits hypha formation and pathogenicity in vitro. Our data indicate that the Ece1 precursor is not required to block premature pore-forming toxicity, but rather to prevent intracellular auto-aggregation of candidalysin sequences.

Toxins are a highly diverse class of harmful molecules that are found throughout all kingdoms of life¹. Mammals produce toxic peptides and proteins such as perforin as components of innate immunity². Bacteria release pore-forming toxins (PFTs) targeting host membranes during infection^{1,3}. Plant pathogenic fungi secrete toxins that induce plant cell death to facilitate nutrient acquisition^{4,5}.

In addition to toxin–antitoxin systems^{6,7}, many peptide and protein toxins are produced as precursors to prevent self-toxicity. All these systems require strictly regulated processing machineries to activate the toxin at the appropriate location and time.

PFTs are particularly well studied in bacteria^{1,8,9}. PFTs initially fold into a water-soluble, generally monomeric structure. Binding of PFT monomers to a specific receptor causes association with target membranes and the formation of multimers, resulting in a toxic pore^{8,10–12}. In contrast to other PFTs, phenol-soluble modulins (PSMs) which are critical determinants of staphylococcal virulence, tend to aggregate into oligomers in receptor-independent processes^{11,12}. Only recently, a similar toxin has been discovered in the opportunistic fungal pathogen *Candida albicans*¹³.

C. albicans causes superficial or life-threatening systemic infections under certain predisposing conditions. Hypha formation is a

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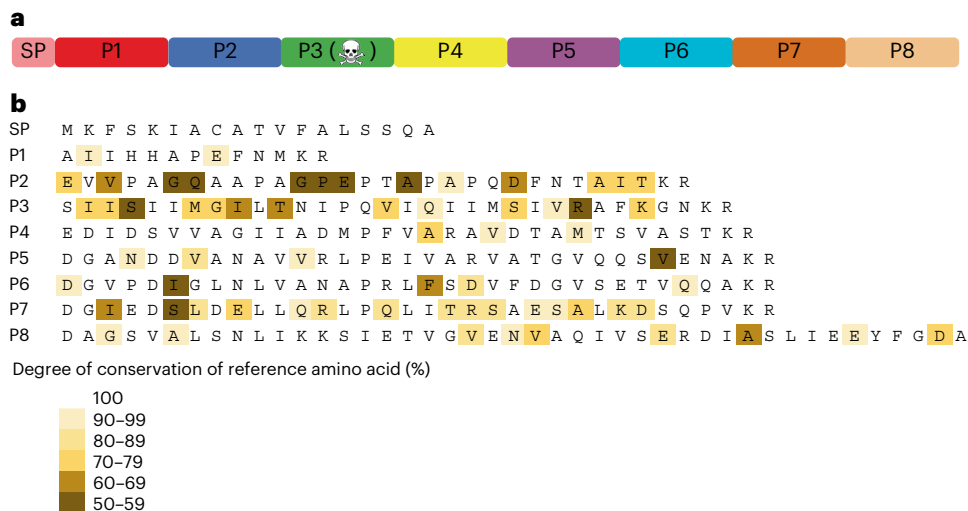


Fig. 1 | Ece1 primary structures and sequences from clinical isolates are conserved. **a**, Scheme of the Ece1 polyprotein primary structure with signal peptide (SP) followed by eight peptides (P1–P8). Skull represents CaL, which is derived from the P3 precursor. **b**, The genomes of 296 *C. albicans* clinical

isolates revealed 46 unique Ece1 protein sequences. A primary alignment of these sequences is shown. The colour code indicates the degree of conservation for each amino acid. While the primary Ece1 structure is highly conserved, most NCEP peptide sequences show moderate variations.

key virulence factor of *C. albicans* and is associated with increased adhesion, invasion and host cell damage¹⁴. Damage of epithelial cells is almost entirely linked to the secretion of the peptide toxin candidalysin (CaL)¹³. Before secretion, CaL is embedded into a polyprotein precursor. This pre-pro-protein (Ece1) consists of a secretion signal peptide, the precursor peptide for CaL (P3) and seven further non-CaL Ece1 peptides (NCEPs), each separated by lysine–arginine residues, which are processing sites for kexin proteases in the Golgi^{13,15}. The precise processing of Ece1 and CaL delivery to the extracellular space is essential for host cell damage¹⁵.

The *ECE1* gene is highly expressed during filamentation¹³ and CaL, secreted by invading hyphae, accumulates in an invasion pocket triggering host responses^{16–18}. While studies have shown that CaL is critical for driving *C. albicans* infections^{13,15–25}, the roles of NCEPs remain unknown.

This study demonstrates that the Ece1 precursor and NCEPs are not required to block premature pore-forming toxicity or to prevent self-toxicity, but rather play a pivotal role in the intracellular folding of Ece1 and the secretion of CaL, most probably by preventing auto-aggregation of CaL sequences within the secretory pathway.

Results

Ece1 is a conserved polyprotein

An alignment of Ece1 protein sequences from 296 *C. albicans* clinical isolates indicates that the unusual structure and specific NCEP sequences are highly conserved (Fig. 1a,b, and Supplementary Fig. 1 and Supplementary Data 1). Therefore, both the Ece1 structure and single NCEPs may be crucial for Ece1 functions in *C. albicans* biology and virulence¹⁶.

CaL-adjacent peptides block host cell toxicity

Pro-domains in toxin precursor systems are well-known mechanisms to prevent premature or self-toxicity^{9,26–29}. Accordingly, NCEPs may prevent premature toxicity to host cells. We investigated whether NCEPs located adjacent to the precursor of CaL (P3) within Ece1 influence CaL toxicity. Thus, we quantified the damage potential of synthetic P2–CaL or P3–P4 fusion peptides against epithelial cells (Extended Data Fig. 1a,b).

Both fusion peptides were unable to cause epithelial damage, whereas synthetic P2 or P4 together with CaL caused an equivalent

level of host cell damage potential as CaL alone (Extended Data Fig. 1a). Therefore, when fused to CaL, peptides flanking the N terminus (P2) or C terminus (P4) of P3 (Extended Data Fig. 1b) are sufficient to prevent cytotoxicity.

NCEPs are not required to prevent self-toxicity

Since fusions of P2 or P4 to CaL or P3 prevent CaL cytotoxicity, we hypothesized that adjacent NCEPs should also prevent integration of CaL into fungal membranes. To address the susceptibility of fungal membranes to CaL, we investigated the direct effect of CaL on *C. albicans*. Neither *C. albicans* yeast or hyphal cells (Fig. 2a,b) nor several other *Candida* species (Extended Data Fig. 2) were susceptible to CaL¹³. In contrast, *Candida* cells (except *C. auris*) (Fig. 2a,b and Extended Data Fig. 2) were highly susceptible to the bee venom peptide melittin (Mel)³⁰. These data suggest that CaL does not damage fungal cells.

Next, we used an indirect approach to evaluate self-toxicity by monitoring the potential of membrane vesicles to interact with CaL. This approach was based on the observation that liposomes can neutralize free bacterial PFTs, thereby protecting host cells^{3,31}. We adapted this approach to investigate whether artificial membrane vesicles (phosphatidylcholine/phosphatidylserine (PCPS) liposomes), extracellular vesicles from human epithelial cells (hEVs) or extracellular vesicles from *C. albicans* (fungal, fEVs) could prevent CaL-induced lysis of epithelial cells (Fig. 2c–e). Strikingly, hEVs and in particular PCPS liposomes, but not fEVs, reduced CaL-induced epithelial damage, indicating that CaL exhibits specificity for human, but not fungal membranes.

To confirm that CaL directly interacts with PCPS liposomes and hEVs, but not fEVs, we used nanoparticle tracking analysis (NTA), which allows monitoring of particle size distribution. Treatment of PCPS liposomes and hEVs, but not of fEVs, with CaL showed substantial alterations in EV size distribution (Fig. 2f–h), indicating that CaL caused particle aggregation or altered membranes of PCPS liposomes and hEVs.

Collectively, these data suggest that CaL neither damages fungal cells nor targets fungal membranes. Consequently, NCEPs and the Ece1 structure are not required to prevent self-toxicity.

Genetic dissection of the role of NCEPs within Ece1

To understand the contribution of each NCEP in the context of CaL-mediated host cell damage, we created a panel of NCEP mutants, each with a modified Ece1 structure (Supplementary Table 1).

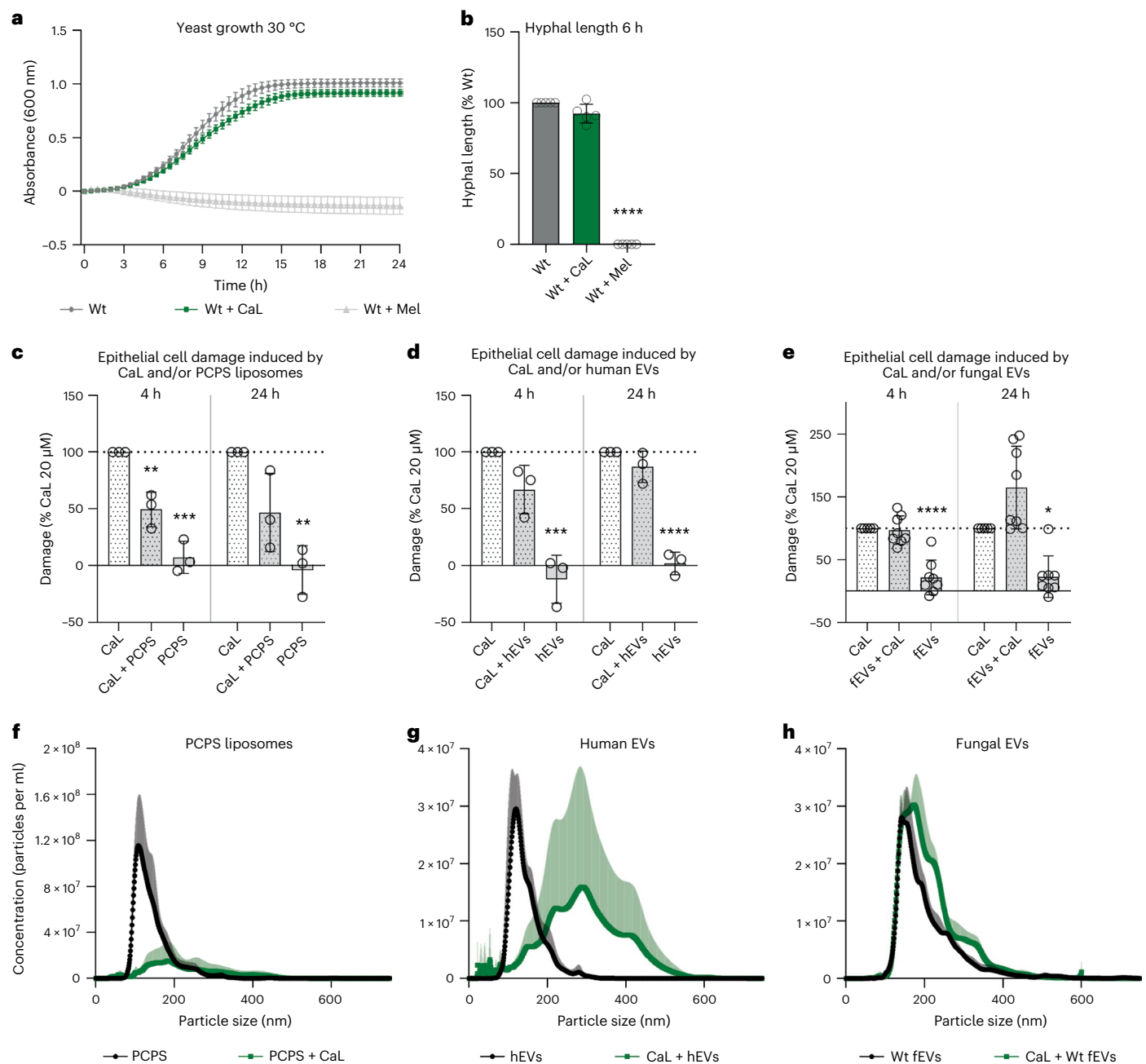


Fig. 2 | **CaL does not integrate into *C. albicans* membranes.** **a**, Yeast growth of Wt *C. albicans* with or without addition of 70 μ M synthetic CaL or synthetic Mel. Yeast growth was monitored for 24 h by measuring the absorbance (600 nm) every 30 min. $n = 6$ biological replicates, mean $\pm$ s.d. **b**, Hyphal length of Wt *C. albicans* with or without addition of 70 μ M synthetic CaL or Mel. Hyphal length was determined microscopically, measuring at least 50 hyphae per replicate after 6 h incubation under hypha-inducing conditions. Data are presented as % Wt hyphal length (mean Wt hyphal length: $122.6 \pm 10.5 \mu$ m, mean $\pm$ s.d.). Statistical analysis was performed with repeated measures (RM) one-way analysis of variance (ANOVA) and a Dunnett's multiple comparisons test relative to the Wt sample; **** $P < 0.0001$, $n = 5$. **c–e**, EV protection assay: damage of TR146 oral epithelial cells was determined after 4 h and 24 h of treatment with artificial liposomes (**c**, PCPS, $n = 3$ biological replicates); oral epithelial cell-derived

extracellular vesicles (**d**, hEVs, $n = 3$ biological replicates); fungal extracellular vesicles (**e**, fEVs, $n = 8$ biological replicates, $n = 5$ for CaL); CaL (20 μ M) or a combination of liposomes/EVs with 20 μ M CaL by quantification of extracellular LDH. Epithelial damage is presented as % CaL-induced damage at 4 h and 24 h post infection, dotted lines represent CaL-induced damage (100%). Statistical analysis for epithelial damage reduction was performed using one-way ANOVA and a Dunnett's multiple comparisons test relative to the respective CaL-treated sample per timepoint; * $P \leq 0.05$, ** $P \leq 0.01$, *** $P \leq 0.001$, **** $P \leq 0.0001$, mean $\pm$ s.d.; more statistical parameters are listed in Source Data Fig. 2. **f–h**, Nanoparticle tracking analysis: size distribution of CaL-treated or untreated PCPS (**f**), hEVs (**g**) and fEVs (**h**). EVs were incubated for 2 h at $37^\circ\text{C} \pm 7^\circ\text{C}$ CaL. Standard deviations are represented by the shaded area in the respective colour; $n = 3$ biological replicates, mean $\pm$ s.d.

First, we created mutants lacking individual NCEPs, including mutants lacking P2, P4, P5, P6, P7 or P8 (Fig. 3a) and studied the impact of these mutations on the damage potential of *C. albicans* (Fig. 3b). A P1 deletion mutant was not created since P1 is probably

required for proper processing of the signal peptide³² and thus essential for co-translation of the Ece1 polypeptide into the endoplasmic reticulum (ER). We also created mutants lacking multiple consecutive peptides (Fig. 3a). Almost all mutants were unable to

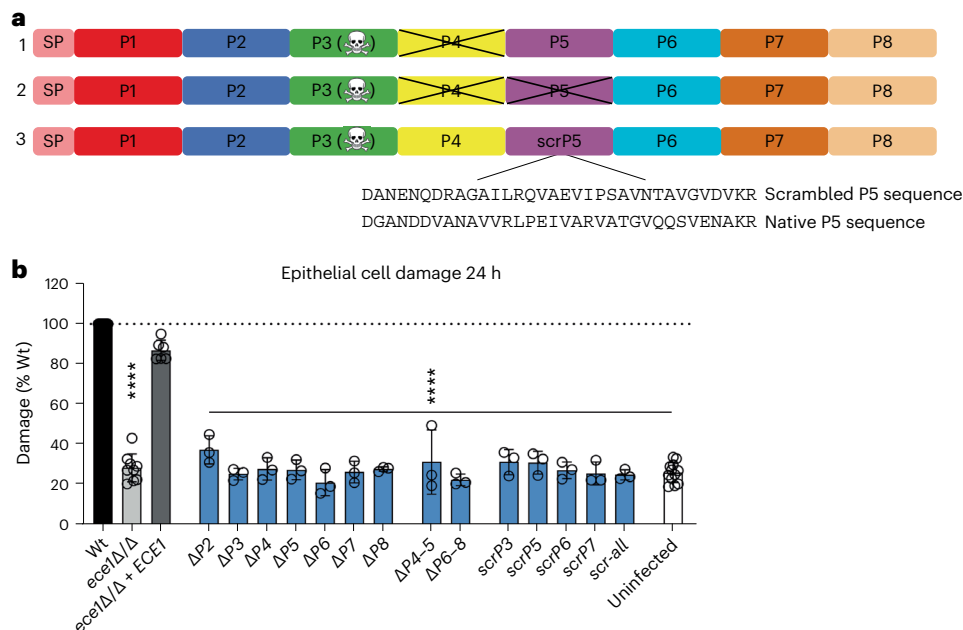


Fig. 3 | *C. albicans* mutants with modified Ece1 sequences exhibit reduced host cell damage potential. a, Classes of NCEP mutants produced in this study: 1: deletion of individual NCEPs within the Ece1 polypeptide; 2: combinatorial deletion of NCEPs; 3: scrambled amino acid sequence of individual (or multiple) NCEPs/peptides. In all mutants, the signal peptide and the lysine–arginine sites are conserved to enable the correct processing by kexin proteases (more details available in Supplementary Table 4). Crosses indicate deletions, the skull represents CaL, which is derived from P3 following Golgi-associated kexin processing. **b**, Damage of TRI46 oral epithelial cells 24 h post infection with Wt

($n = 12$), *ece1Δ/Δ* ($n = 9$), *ece1Δ/Δ + ECE1* ($n = 6$) complemented strain, a mutant strain lacking the CaL-encoding sequence only ($\Delta P3$), NCEP mutant strains or scrambled peptide mutant strains ($n = 3$). Data are presented as % damage relative to the Wt, dotted line represents Wt-induced damage (100%). Host cell damage was assessed by quantification of extracellular LDH. Statistical analysis was performed using one-way ANOVA and a Dunnett's multiple comparisons test relative to the raw absorbance values of the Wt sample (**** $P < 0.0001$; NS, not significant, $P = 0.0555$; $n = 12$ biological replicates, uninfected sample; mean $\pm$ s.d.).

damage epithelial cells (Fig. 3b) comparable to the non-damaging $\Delta P3$ mutant¹³.

We next produced mutants that retained the same overall size but harboured scrambled peptide sequences (Fig. 3a), including scrambled P5 (*scrP5*), *scrP6*, *scrP7* and all peptides scrambled (*scr-all*). All mutants were unable to damage epithelial cells (Fig. 3b).

To determine why these mutants exhibited reduced damage potential, we analysed the phenotypes of the NCEP mutants. As hypha formation is inextricably linked with CaL damage¹⁶, we monitored the growth and filamentation of all mutants. While yeast growth was unaffected (Fig. 4a), filamentation of several mutants was significantly attenuated at 6 h (Fig. 4b): deletions of P4–P5 or P6–8 severely affected hypha formation, while deletion of peptides P2, P5, P6, P7 or P8 led to moderate hyphal reduction. All NCEP mutants did not revert to yeast morphology but grew as hyphae in the initial phase until 24 h (Fig. 4c and Extended Data Fig. 5a). Only the *scr-all* mutant grew predominantly in the yeast morphology. Notably, mutants lacking P3 (*ece1Δ/Δ* and $\Delta P3$) exhibited normal hypha formation, confirming that hypha formation per se is not sufficient to damage host cells and that deletion of the CaL-encoding sequence has no effect on filamentation¹³ (Fig. 4b,c). In conclusion, these data suggest that impaired hypha formation in the NCEP mutants probably contributes to the reduced host cell damage potential¹⁶.

NCEP mutations trigger an unfolded protein response

Next, we reasoned why NCEP mutants exhibited impaired filamentation. For example, a dysfunctional secretory pathway may have caused reduced hyphal extension³³. We postulated that *ECE1* alterations in NCEP mutants may result in a modified structure of the Ece1 polypeptide, which may lead to misfolding of Ece1 in the ER and insufficient transport to the Golgi, where Ece1 processing occurs (Extended

Data Fig. 3). If so, this would trigger ER stress and an unfolded protein response (UPR), which in turn would impair hypha formation^{33–35}.

To confirm that activation of the UPR correlated with altered morphology, we treated *C. albicans* with tunicamycin which induces ER stress and a UPR by inhibiting N-linked glycan biosynthesis^{33–35}. Tunicamycin treatment affected yeast growth (Extended Data Fig. 4a,b) and inhibited hypha formation (Extended Data Fig. 4c,d), confirming previous reports^{35,36}.

C. albicans relies on inositol-requiring enzyme 1 (Ire1) for sensing ER stress and activating the UPR³³. Hac1, a transcriptional activator of UPR genes, is the downstream target of Ire1. Ire1 induces splicing of *HAC1* messenger (m)RNA during ER stress and UPR induction³⁷. This results in Hac1 translocation into the nucleus and the activation of UPR-responsive genes³⁸.

To measure ER stress in the NCEP mutants, the induction of the UPR was monitored by analysing *HAC1* mRNA splicing³⁹ by PCR amplification of *HAC1* complementary (c)DNA^{33,35}. Full length *HAC1* cDNA yielded an amplification product of 116 bp, whereas spliced *HAC1* mRNA yielded a product of 97 bp.

We hypothesized that the UPR, and thus *HAC1* mRNA splicing, would be induced during hypha formation (when *ECE1* is expressed), but not during yeast growth (no *ECE1* expression) of NCEP mutants. Accordingly, we monitored *HAC1* mRNA splicing 3 h after hyphal induction. As expected, *HAC1* mRNA splicing occurred in tunicamycin-treated yeast and hyphal cells. Wild-type (Wt) *C. albicans*, *ece1Δ/Δ* and mutants lacking the CaL precursor P3 did not show *HAC1* mRNA splicing (Fig. 5b and Supplementary Fig. 2), indicating the absence of ER stress in strains that either lack CaL or express CaL in its native NCEP environment. In contrast, mutants $\Delta P8$, $\Delta P4-5$, $\Delta P6-8$ and mutants with scrambled peptide sequences demonstrated *HAC1* mRNA splicing under hypha-inducing conditions, indicating ER stress (Fig. 5b and

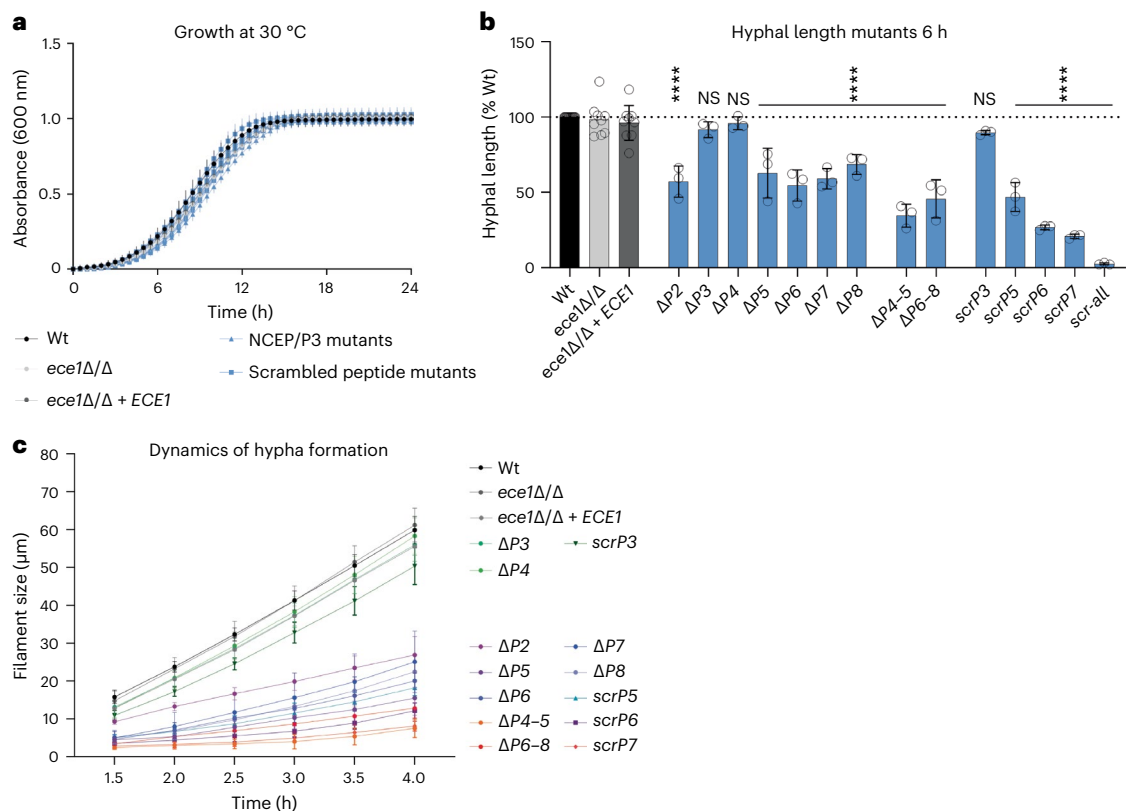


Fig. 4 | Mutants with Ece1 sequence variation exhibit dysfunctional hypha formation. **a**, Yeast growth of Wt ($n = 12$ biological replicates), *ece1Δ/Δ* ($n = 9$), *ece1Δ/Δ + ECE1* ($n = 7$) complemented strain, $\Delta P3$, NCEP mutants and scrambled peptide mutants ($n = 3$). Yeast growth was monitored for 24 h by measuring the absorbance (600 nm) every 30 min at 30 °C. Data were normalized following subtraction of the first measurement (baseline); mean $\pm$ s.d. **b**, Hyphal length of indicated *C. albicans* strains after 6 h of hypha induction. Data are presented as % Wt hyphal length (mean hyphal length of the Wt: 122.0 ± 30.5 μm). Dotted line represents the length of Wt hyphae. Hyphal length was determined

microscopically by measuring at least 50 hyphae per replicate. Statistical analysis was performed using one-way ANOVA and a Dunnett's multiple comparisons test relative to the Wt sample. **** $P < 0.0001$; Wt $n = 18$, *ece1Δ/Δ* and *ece1Δ/Δ + ECE1* ($n = 9$), NCEP mutants ($n = 3$), mean $\pm$ s.d.; more statistical parameters are listed in Source Data Fig. 4. **c**, Dynamics of hypha formation of Wt, *ece1Δ/Δ*, *ece1Δ/Δ + ECE1* complemented strain, $\Delta P3$, NCEP deletion mutants and scrambled peptide mutants (except *scr-all*). Filament length was measured from 1.5 h to 4 h every 30 min on plastic at 37 °C and 5% CO₂ ($n = 3$, ≥ 20 hyphae per replicate were tracked over time, except one replicate of $\Delta P8$ $n = 19$ hyphae, mean $\pm$ s.d.).

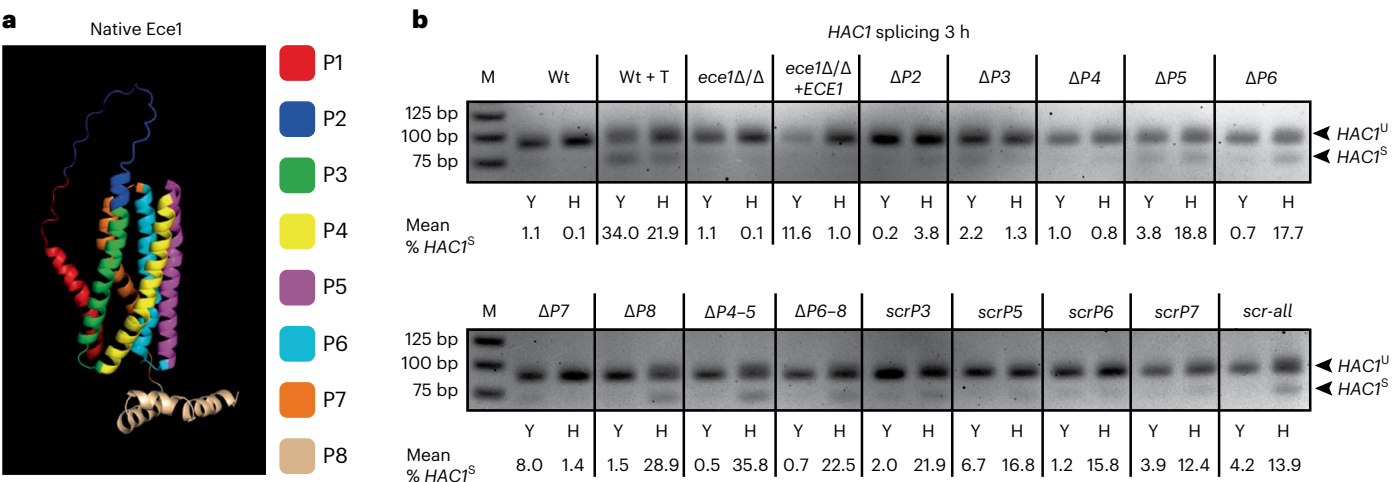


Fig. 5 | ER stress in NCEP mutants activates the UPR. **a**, AlphaFold-predicted peptide structure of native Ece1. **b**, Analysis of HAC1 mRNA splicing as a read-out for ER stress and UPR induction. Activation of the UPR was determined by visualization of HAC1 mRNA splicing by RT-qPCR after culturing Wt, *ece1Δ/Δ*, *ece1Δ/Δ + ECE1* complemented strain, $\Delta P3$, NCEP deletion mutants and scrambled NCEP mutant strains for 3 h under yeast- or hypha-inducing conditions. A band at 97 bp shows HAC1 mRNA splicing, indicating ER stress and UPR activation.

Y, yeast growth conditions (30 °C, 180 r.p.m. shaking); H, hyphal growth conditions (37 °C, 5% CO₂, plastic surface); positive control: Wt treated with tunicamycin (T) ($2 \mu\text{g ml}^{-1}$ (Wt + T)). One representative replicate from three biological repeats is shown. Quantification of HAC1 mRNA splicing is presented as the percentage of spliced HAC1 mRNA from total HAC1 mRNA level (mean of three biological replicates, quantified with BioID from Vilber Lourmat). Graphs presenting means $\pm$ s.d. and statistics can be found in Supplementary Fig. 2.

Supplementary Fig. 2). These data suggest that minor alterations of the primary Ece1 structure can lead to a UPR, which in turn affects hypha formation.

We further hypothesized that the activated UPR would ameliorate ER stress at a certain threshold due to the removal of misfolded Ece1 and that, accordingly, impaired hypha formation in the majority of NCEP mutants may be a dynamic phenomenon. Indeed, NCEP mutants continued to grow as hyphae at later timepoints (Extended Data Fig. 5a). Thus, we monitored *HAC1* mRNA splicing in NCEP mutants over time to investigate the course of UPR activation. At 20 min (immediate), a global UPR response was observed in yeast and hyphal conditions, probably due to accelerated growth associated with the switch from pre-culture to main culture conditions (Extended Data Fig. 5b). At 1 h, 2 h and 3 h (early), UPR activation was observed only in hyphal conditions (Fig. 5b, Extended Data Fig. 5c and Supplementary Fig. 2). At 24 h (late) the UPR activation was nearly abolished for all NCEP mutants and the tunicamycin-treated hypha condition (Extended Data Fig. 5b and Supplementary Fig. 2).

In conclusion, ER stress induced by misfolded Ece1 is a dynamic process depending on the type of modification to Ece1 and the importance of distinct NCEPs to prevent misfolding. ER stress was ameliorated over time as a result of the UPR, indicated by the absence of *HAC1* mRNA splicing and the recovery of hyphal growth at later stages.

Finally, since altering the Ece1 structure induces pleiotropic cellular effects, we hypothesized that other processes required for CaL export may also be negatively affected, causing reduced CaL secretion from NCEP mutants. Therefore, we used an anti-CaL nanobody¹⁶ to detect CaL antigen in the vicinity of NCEP mutant hyphae, and nano-liquid chromatography–mass tandem spectrometry (LC–MS/MS) to quantify CaL secretion (Extended Data Fig. 6, Supplementary Data 2 and Supplementary Fig. 3). As expected, CaL antigens were not detected in mutants lacking native CaL sequences (*ece1Δ/Δ*, $\Delta P3$, *scrP3* and *scr-all*) (Extended Data Fig. 6a,b, Supplementary Data 2 and Supplementary Fig. 3). In all other NCEP mutants, several CaL secretion parameters were negatively affected, including the amount of CaL antigen on hyphae, the number of CaL-positive hyphae and CaL secretion (Extended Data Fig. 6a,b, Supplementary Data 2 and Supplementary Fig. 3). Impaired CaL secretion observed in NCEP mutants correlated with reduced epithelial damage (Fig. 3b), reduced hypha formation (Fig. 4b) and diminished invasion (Extended Data Fig. 6c).

The Ece1 polypeptide structure may prevent toxin aggregation

The UPR induced by an altered Ece1 structure may be explained by the unusual hydrophobicity of CaL (P3) sequences. Improper or inefficient shielding of the hydrophobic regions may be associated with auto-aggregation within the secretory pathway. In this case, it would be expected that isolated CaL peptides would aggregate in vitro, similar to staphylococcal PSMs which tend to aggregate in aqueous solutions¹¹.

We used dynamic light scattering (DLS) to investigate concentration-dependent aggregation of CaL peptides in solution. At CaL concentrations above 1 μ M, aggregation was significant (Extended Data Fig. 7).

We therefore hypothesized that a major function of the Ece1 polypeptide is to prevent auto-aggregation of CaL sequences. This view is supported by the predicted three-dimensional (3D) structure of Ece1 using AlphaFold (Fig. 5a), which suggests that peptides P4–P8 form α -helical structures, with peptides P4–P7 located closely and in parallel to the α -helical P3, the precursor of CaL. Consequently, modifications to the Ece1 polypeptide structure (Supplementary Data 3) may lead to an increased likelihood of aggregation of CaL sequences, Ece1 misfolding and UPR induction within the ER.

These data suggest that NCEPs are required to prevent auto-aggregation of CaL sequences while concomitantly permitting efficient Ece1 processing and escort of CaL to the extracellular space. This leads to two testable hypotheses; (1) multiple CaL precursors (P3)

in a modified Ece1 protein would be more detrimental than the removal of NCEPs in the presence of one copy of P3 and (2) the removal of NCEPs would have a minor effect if CaL sequences are absent. In the first case, a strong UPR response would be predicted due to excessive aggregation of multiple CaL sequences that are insufficiently protected by the remaining NCEPs. In the second case, NCEPs would not be required for protection as CaL sequences have been removed and aggregation would not occur. Consequently, removal of NCEP peptides would show a Wt-like phenotype in the absence of P3.

To test the first hypothesis, we created an Ece1 mutant containing three copies of P3 by replacing P5 and P7 with two additional P3 sequences (*TripleP3*) (Extended Data Fig. 8a). As predicted, the *TripleP3* mutant exhibited strongly reduced host cell damage (Extended Data Fig. 8b), a severe reduction in hypha formation (Extended Data Fig. 8c–e), a diminished ability to secrete CaL (Extended Data Fig. 8h,i, Supplementary Data 2 and Supplementary Fig. 3) and activation of the strongest UPR among all NCEP mutants (Extended Data Fig. 8f). Consequently, intracellular accumulation of the precursor polypeptide of the *TripleP3* mutant was observed (Extended Data Fig. 8g). To test the second hypothesis, two mutants with P3 and P5 ($\Delta P3 + 5$) or P3 and P7 ($\Delta P3 + 7$) deleted were constructed (Extended Data Fig. 8a). As expected, all mutants lacking the CaL-encoding sequence did not secrete the toxin as shown by immunohistochemistry and LC–MS/MS analysis (Extended Data Fig. 8h,i, Supplementary Data 2 and Supplementary Fig. 3). The $\Delta P3 + 5$ and $\Delta P3 + 7$ mutants did not damage host cells, exhibited Wt-like hypha formation and did not exhibit a UPR, similar to the $\Delta P3$ mutant (Extended Data Fig. 8b–f).

These data suggest that NCEPs play a fundamental role in *C. albicans* biology by preventing auto-aggregation of CaL sequences, thereby ensuring proper Ece1 folding, processing, normal hyphal growth and efficient CaL secretion.

NCEPs interact with CaL

We hypothesized that, following Ece1 folding and processing, NCEPs should still associate with CaL after secretion to prevent premature extracellular aggregation but must ultimately dissociate from CaL to permit toxin function. Our previous data showed that several NCEPs are digested into subfragments¹³. Notably, P5 and P7 subfragments (P5a, P5b, P7a and P7b) were observed in supernatants of hyphae cultures¹³.

Therefore, using DLS we investigated the ability of NCEPs and P5/P7 subfragments to modify CaL aggregation in solution. Except for P6, none of the NCEPs exhibited aggregation at a concentration of 100 μ M, while CaL aggregated at a concentration above 1 μ M (Fig. 6a and Extended Data Fig. 7). Equimolar mixtures of CaL and P7, but not P7a or P7b, led to a significant increase in count rates, suggesting that P7 can directly bind to CaL and processing of P7 into subfragments abolished the P7 effect on CaL. Notably, P1, P2, P4 and P8 resulted in decreased CaL aggregation, while scrambled versions of P1, P2 and P4 had no effect (Fig. 6a,b).

We further hypothesized that the interaction of NCEPs with CaL may modulate CaL-induced host membrane damage once the toxin is released into the extracellular space. To investigate whether CaL-induced membrane permeabilization is altered by NCEPs, we created tethered 1,2-dioleoyl-*sn*-glycero-3-phosphocholine (DOPC) membranes and determined their electrophysiological integrity after co-incubation of CaL with selected NCEPs and their fragments. The addition of CaL caused membrane permeabilization beginning at a toxin concentration of 4 μ M (Fig. 6c). In contrast, the tested NCEP subfragments alone did not induce pore formation (as exemplified by P7a, Fig. 6c). Notably, co-incubation of CaL with equimolar concentrations of P5 and P7 reduced CaL-induced membrane permeabilization by more than 90%. Subfragments P7a and (to a lesser extent) P7b also reduced CaL-induced membrane permeabilization, while P5a and P5b had no effect (Fig. 6c).

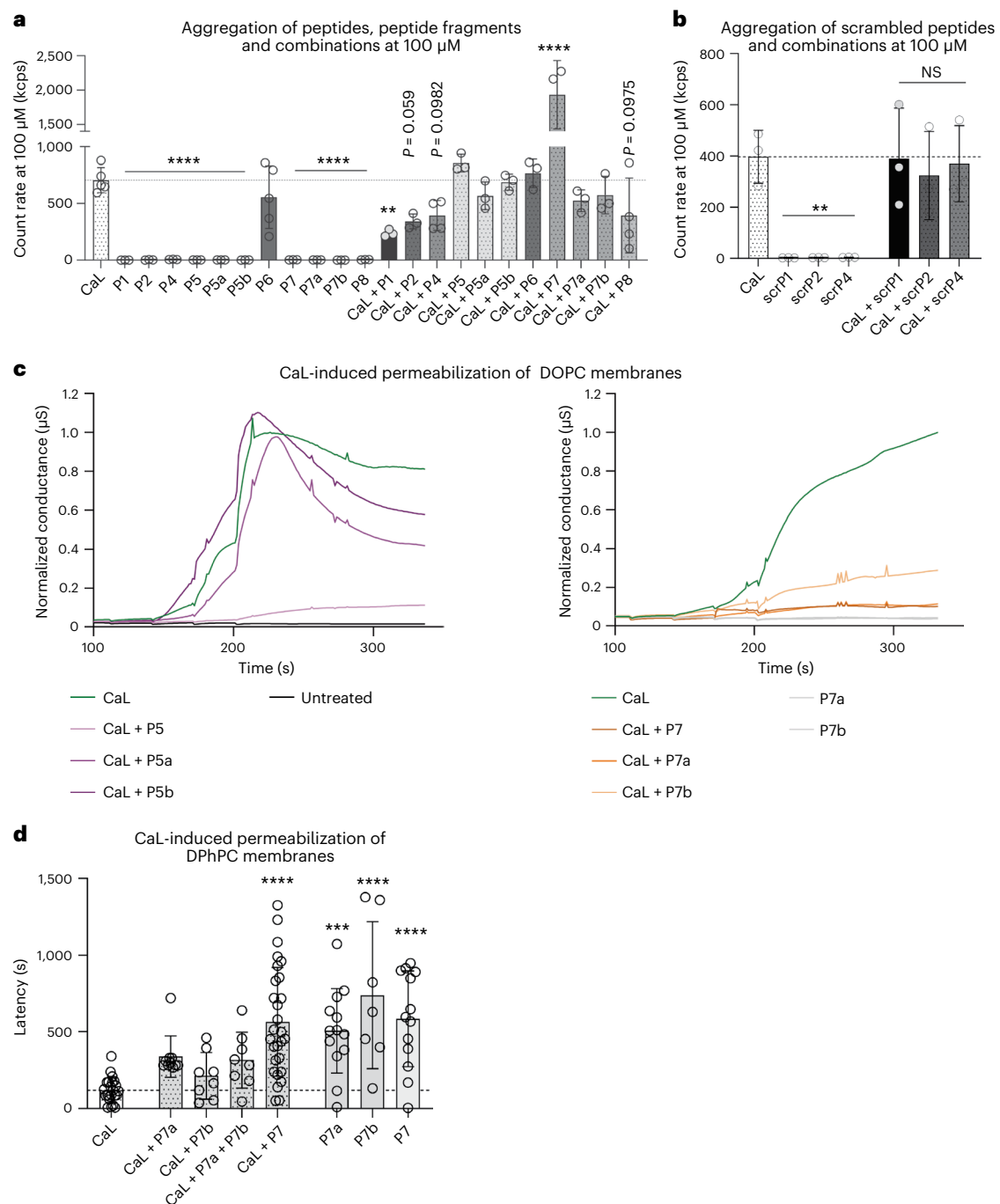


Fig. 6 | CaL aggregation and CaL-induced membrane permeabilization is altered by NCEPs and their subfragments. a, Count rate at 100 μM CaL compared to NCEPs at 100 μM and 100 μM CaL + 100 μM of the respective NCEPs. Increased count rates represent a strong gain in shape, concentration and/or size of aggregates detected by the Zetasizer. Each bar represents the mean value of a titration experiment. Statistical analysis was performed using one-way ANOVA and a Dunnett's multiple comparisons test relative to the CaL-treated sample; mean $\pm$ s.d.; ** P = 0.0051, **** P < 0.0001; exact P values are listed in the source data; n = 3 independent experiments, CaL and P6 n = 5, CaL + P4 and CaL + P8 n = 4. **b**, Count rate at 100 μM CaL compared to scrambled (scr) NCEPs at 100 μM and 100 μM CaL + 100 μM of the respective scr NCEPs. Statistical analysis was performed using one-way ANOVA and a Dunnett's multiple comparisons test relative to the CaL-treated sample, n = 3 experiments; mean $\pm$ s.d.; ** P < 0.01; more statistical parameters are listed in Source Data Fig. 6. **c**, Alterations in normalized conductance n_{Gm} of tethered DOPC lipid membranes after addition of CaL, CaL with an equimolar amount of P5/P7, CaL with an equimolar

amount of P5a/P5b and P7a/P7b subfragments. Concentrations: 20 min, 0.5 μM ; 50 min, 1 μM ; 80 min, 1.5 μM ; 110 min, 2 μM ; 140 min, 4 μM ; 170 min, 6 μM ; 200 min, 8 μM . Data show representative curves from three individual experiments. Additional controls for P5, P5a, P5b, P7 and P7b showed no pore formation. Data were normalized to the highest CaL-induced conductance value measured. **d**, 10 μM CaL-induced membrane permeabilization was monitored with Orbit e16 using DPhPC lipid membrane bilayers (121 ± 80 s; n = 23). The latency time from peptide addition until membrane permeabilization was quantified. To monitor the effect of P7, P7a and P7b, an equimolar amount of each peptide was pre-incubated with CaL. A significant delay in membrane permeabilization was observed when CaL was pre-incubated with P7 (average latency time = 566 ± 354 s). Statistical analysis was performed using one-way ANOVA and a Dunnett's multiple comparisons test relative to the CaL-treated sample; **** P = 0.0006; **** P < 0.0001; more statistical parameters are listed in Source Data Fig. 6; CaL + P7a n = 10 independent experiments, CaL + P7b and CaL + P7a + P7b n = 8, CaL + P7 n = 28, P7a and P7 n = 13, P7b n = 7; mean $\pm$ s.d.

Since P7 and its subfragments P7a and P7b exhibited the most efficient reduction in DOPC membrane permeabilization, we continued our analysis using Orbit el6 technology to quantify the kinetics of membrane permeabilization. Artificial lipid bilayers were created using a zwitterionic lipid (1,2-diphytanoyl-*sn*-glycero-3-phosphocholine (DPhPC)) and current changes across the bilayer were monitored following the addition of CaL with P7, P7a and P7b, and latency (defined as time elapsed between the addition of CaL/NCEP complexes and bilayer permeabilization) was quantified. Addition of CaL generated an average latency time of 119 ± 79 s (Fig. 6d). In contrast, pre-incubating CaL with equimolar concentrations of P7 reduced membrane permeabilization by 4.7 times. Pre-incubation of CaL with equimolar P7a, P7b or combined P7a and P7b showed a non-significant reduction in membrane permeabilization, while latency values observed by combining CaL with equimolar concentrations of P7a and P7b were similar to those obtained using P7a alone (Fig. 6d). Notably, P7, P7a and P7b alone did not induce membrane permeabilization (Fig. 6d).

The combined biophysical data suggest that NCEPs are critical in interacting with and modifying the physical properties of CaL, which in turn may prevent auto-aggregation as Ece1 is trafficked through the secretory pathway.

Discussion

The production of pore-forming toxins (PFTs) by microbes requires sophisticated mechanisms to prevent self-toxicity. These mechanisms include a diverse array of toxin–antitoxin systems and defined protein precursors for protein and peptide toxins, respectively^{6–8,27}. Following attachment, PFTs insert into their target membrane to form transmembrane pores causing membrane permeabilization. These peptides have amphipathic characteristics that are essential for this process: hydrophobic regions interact with membrane lipids, while hydrophilic regions either interact with the phospholipid head groups or face the lumen of the pore⁴⁰. The hydrophobicity of peptide toxins poses an additional deleterious risk to the microbes that produce them due to possible hydrophobic interactions with self-proteins and membranes. The mechanisms used to mitigate these risks are poorly characterized. In many cases, only binding to specific receptors located on the surface of target cells leads to toxin accumulation and subsequent oligo- or polymerization⁸. The PSMs of staphylococci aggregate readily^{11,12}, while dedicated ABC transporter systems prevent intracellular accumulation of PSMs¹¹.

CaL¹³ shares similarities with PFTs such as PSMs and Mel. However, in contrast to all known PFTs, the precursor of CaL is located within a complex polyprotein¹⁵. The processing of Ece1 resembles the maturation of yeast killer toxins, in which a pre-pro-toxin is a substrate for kexin-mediated trimming in the Golgi²⁷. However, uniquely, CaL manifests as a single monomer embedded in a pro-toxin consisting of several peptides. Numerous secreted proteins and peptides of eukaryotes are translated as precursors that require post-translational site-specific proteolysis by pro-protein convertases such as kexins⁴¹. In many cases, removal of a pro-domain from the N terminus of the protein is sufficient to activate the corresponding effector protein or peptide, for example, Mel⁴².

Here we show that the primary Ece1 structure is highly conserved among clinical isolates of *C. albicans* (Fig. 1, Supplementary Fig. 1 and Supplementary Data 1), suggesting that NCEPs play an important role in *C. albicans* biology. On the basis of extensive PFT literature, we initially hypothesized that the function of NCEPs might be to prevent self-toxicity in the fungus. However, *C. albicans* and other *Candida* species were not susceptible to CaL (Fig. 2a,b and Extended Data Fig. 2). Furthermore, artificial membrane vesicles and EVs from human cells, but not fungal EVs, protect host cells against CaL-induced damage (Fig. 2c–e), indicating specificity of the toxin for human host but not for fungal membranes, which was confirmed by NTA (Fig. 2f–h). In addition, single adjacent NCEPs fused to CaL are sufficient to prevent

host-directed toxicity (Extended Data Fig. 1). Therefore, the main function of NCEPs and the Ece1 structure is not to prevent self-toxicity.

Mutants lacking either single or multiple NCEP sequences, or that contain scrambled NCEP sequences, were unable to cause wild-type levels of host cell damage (Fig. 3b). We found that the majority of NCEP mutants were impaired for hypha production (Fig. 4b,c and Extended Data Fig. 5a), an essential prerequisite to deliver CaL and to damage host cell membranes¹⁶.

We hypothesized that impaired hypha formation may be associated with a dysfunctional Ece1 polyprotein. Given the conservation of the primary Ece1 structure and the moderate variations of NCEP sequences between clinical isolates, we proposed that even minor modification of the Ece1 polyprotein may lead to misfolding, ER stress and a dysfunctional secretion machinery. Indeed, almost all modifications of Ece1 investigated in this study, except removal of P4, led to a UPR. The UPR of several NCEP mutants was comparable to treatment with tunicamycin, a well-known UPR inducer^{33,35} (Fig. 5b) and inhibitor of *C. albicans* hypha formation^{33,35,36}, confirming our observations (Extended Data Fig. 4a–d). Therefore, we concluded that the reduced host cell damage of NCEP mutants was due to Ece1 misfolding, which triggers a UPR and reduced filament formation with concomitant Ece1 degradation, resulting in lower levels of CaL secretion.

However, the question remained: why does CaL need to be embedded in such a complicated and sensitive structure that requires high conservation? The most likely explanation lies in the extreme hydrophobic nature of CaL sequences, which leads to physical aggregation and clumping of unprotected CaL molecules in aqueous solution (Extended Data Fig. 7). Clearly, such auto-aggregations need to be prevented from initial (co-)translation of Ece1, transport along the secretory pathway, to the delivery of CaL into the extracellular space, and our data support the view that this is probably the main function of NCEPs.

The predicted 3D structure of the Ece1 polyprotein using AlphaFold⁴³ (Fig. 5a and Supplementary Data 3) suggests that α -helical P3, the precursor of CaL, is located closely and in parallel to other α -helical NCEPs, potentially preventing putative intermolecular interaction with other proximate P3 peptides in other Ece1 polyproteins. Thus, NCEPs probably prevent auto-aggregation of CaL sequences, and facilitate its transport through the secretory pathway and its release into the extracellular environment. This would imply that secreted CaL is associated with NCEPs. Indeed, our data suggest that NCEPs may guide the delivery of CaL to the external environment. An intriguing finding from our previous work was that P7 subfragments (P7a and P7b) are secreted from *C. albicans* hyphae together with CaL^{13,15}. Here we show that while P7 can prevent CaL permeabilization of artificial membranes, this ability of P7 is modified when processed into P7a and P7b (Fig. 6c). Thus, we propose that P7 is a fundamentally important NCEP that contributes to the prevention of CaL auto-aggregation within the fungal cell and helps to guide the toxin through the secretory pathway to the hyphal surface, where P7 is processed to permit safe secretion of CaL.

It should be noted that not all NCEPs may be equally important for the 3D structure of Ece1. For example, we could not detect a UPR for the mutant lacking P4. This mutant demonstrated normal hypha formation (Fig. 4b,c and Extended Data Fig. 5a) and reduced CaL antigen exposure on hyphal surfaces (Extended Data Fig. 6a). The CaL nanobody does not only detect free CaL, but also CaL antigens in association with other Ece1 peptides. Indeed, the $\Delta P4$ mutant secretes very low levels of native CaL as observed by LC–MS/MS (Supplementary Data 2).

It is conceivable that P4 has functions unrelated to the prevention of Ece1 misfolding. For example, processing of the C-terminal KR site of P3 may be dysfunctional in this mutant. Amino acid sequences C-terminal and N-terminal to the KR processing sites are known to be important for efficient processing by Kex2 (ref. 44). This would be similar to the reduced damage potential reported in *C. albicans* strain 529L⁴⁵, which

could be explained by sequence variations next to the N-terminal KR processing site of P3 causing improper processing. Consequently, CaL may still be bound to its adjacent peptide and such fusion peptide variants are unable to cause damage in epithelial cells (Extended Data Fig. 1a). Likewise, removal of the N-terminal or C-terminal KR sites of P3 renders *C. albicans* unable to induce epithelial damage¹⁵.

In summary, we demonstrate that *C. albicans* Ece1 has an exceptional and conserved polypeptide structure comprising several peptides that play vital roles in the folding and processing of intracellular Ece1, probably preventing CaL auto-aggregation and guiding CaL through the secretory pathway to permit safe secretion of the toxin to the extracellular space. The evolutionary conservation of such a defined structure must be associated with a high selective pressure and thus indicates the fundamental importance of CaL for the lifestyle of *C. albicans* and may explain why this fungus is the most successful pathogenic *Candida* species.

Methods

Identification of *C. albicans* ECE1 haplotypes and Ece1 variants

Single nucleotide polymorphisms in the *ECE1* gene of 296 genome-sequenced *C. albicans* isolates⁴⁶: CdE, MEB and CM, (Supplementary Data 1) were retrieved and used as input in PHASE (v.2.1.1)^{47,48}. Fifty-three *C. albicans* ECE1 haplotypes were identified (presented in Supplementary Data 1), whose translation enabled the identification of 46 *C. albicans* Ece1 variants. The frequency of the predominant amino acid at each position across the 46 Ece1 variants was calculated and is summarized in Fig. 1, and Supplementary Fig. 1 additionally displays the amino acid variants at each position along Ece1.

Mammalian cell culture

Human oral epithelial cells (TR146) were used throughout this study. The cell line was purchased from the European Collection of Authenticated Cell Cultures⁴⁹ (Sigma-Aldrich, 10032305-1VL) and cultured at 37 °C and 5% CO₂ in a humidified incubator in Dulbecco's modified Eagle medium/Nutrient Mixture F-12 (DMEM/F12, Gibco, 11320074) supplemented with 10% (v/v) heat-inactivated (30 min at 56 °C) fetal bovine serum (Bio&Sell). Confluent TR146 cells were washed once with phosphate-buffered saline (PBS) and subsequent experiments were carried out in serum-free medium. The cell line authenticity was lastly verified by short tandem repeat analysis (Eurofins, 17 February 2023).

Synthetic peptides

Synthetic CaL, peptide 1 (P1), P2, P4, P5, P5a, P5b, P6, P7, P7a, P7b, P8, P2-CaL, P3–P4, scrambled (scr) peptides scrP1, scrP2 and scrP4 were purchased from Peptide Synthetics. Mel was purchased from ProteoGenix. Lyophilised peptide was dissolved to a concentration of 1.4 mM in the appropriate volume of water and stored at –20 °C. Peptide sequences are listed in Supplementary Table 2.

General cultivation of *Candida* strains and *Escherichia coli*

***Candida* species and strains.** *C. albicans* strains used in this study are shown in Supplementary Table 1. *C. glabrata* (ATCC2001), *C. tropicalis* (DSM4959), *C. parapsilosis* (GAI)⁵⁰, *C. auris* (B8441)⁵¹ and *C. haemulonii* (CBS5149T) were received from the American Type Culture Collection (ATCC), the German Collection of Microorganisms and Cell Cultures (DSMZ), the Fungal Biodiversity Centre (CBS), the Mycotic Diseases Branch of the US Centers for Disease Control and Prevention, or kindly provided by A. Gacser, University of Szeged, Hungary.

C. albicans strains and *Candida* species were cultured overnight in liquid YPD medium (1% (w/v) yeast extract (Serva, 24540.03), 2% (w/v) peptone (Bacto, 211677-4), 2% (w/v) dextrose (Roth, 6887.2)) at 30 °C, shaking at 180 r.p.m. To adjust the cell concentration before experiments, the overnight cultures were washed twice with PBS, counted and diluted as indicated for the respective experiment.

E. coli was cultured in LB agar plates or liquid medium (180 r.p.m. shaking) at 37 °C. Where necessary, the medium was supplemented with 50 µg ml^{–1} ampicillin (Roth, HP62.2).

Growth assays

To assess *C. albicans* yeast growth, cells were adjusted to an optical density (OD)₆₀₀ of 0.1 in YPD and growth was monitored by measuring the absorbance at 600 nm over a time period of 24 h at 30 °C in an Infinite 200 PRO ELISA Reader (Tecan, software Tecan i-control v.2.0.10.0), with measurements conducted every 30 min after 30 s of shaking before each measurement. At least three biological replicates were performed.

To determine hypha formation, 3 × 10⁴ *C. albicans* cells per well were inoculated in 1 ml of RPMI 1640 medium (Gibco 21875091) into 24-well plastic plates and incubated for 6 h at 37 °C and 5% CO₂. For determination of late-stage filamentation size, 200 *C. albicans* cells per well were inoculated in 1 ml of RPMI 1640 medium and incubated for 24 h at 37 °C at 5% CO₂. Samples were fixed using formaldehyde (Roth, 4979.1). At least 50 hyphae or least 20 microcolonies per sample were measured using a Zeiss Axio Vert microscope and Zen lite v.2.4 software (ZEISS). Since at 24 h single hyphae were not measurable due to the formation of microcolonies, the diameter of microcolonies was measured to evaluate the filamentation behaviour.

To determine the influence of CaL and Mel on yeast and hyphal growth of *C. albicans*, yeast and hyphal growth conditions were analysed as described above with the addition of 70 µM synthetic CaL and Mel.

Growth of *C. albicans* was assessed at 30 °C and 37 °C in the presence of 2 µg ml^{–1} tunicamycin (Sigma-Aldrich, T7765-5MG).

Host cell damage assay

To analyse the *C. albicans*-induced damage of TR146 cells, the activity of lactate dehydrogenase in exhausted culture medium was measured 24 h postinfection using the cytotoxicity detection kit (Roche, 11644793001) according to the manufacturer's instructions. Absorbance was measured with an Infinite 200 PRO ELISA Reader (Tecan, software Tecan i-control v.2.0.10.0). Briefly, 2 × 10⁴ TR146 cells per well were seeded in a 96-well plate in serum-free DMEM/F12 (Gibco, 11320074) and grown to confluency at 37 °C and 5% CO₂. Subsequently, TR146 cells were infected with *C. albicans* at a multiplicity of infection of 1 and incubated for 24 h at 37 °C and 5% CO₂.

To detect the epithelial damage induced by synthetic peptides, the respective peptides were diluted to the appropriate concentration in serum-free DMEM/F12 and applied to TR146 cells, the latter seeded as described. Cells were incubated at 37 °C and 5% CO₂ for 4 h and/or 24 h.

Preparation of lipid liposomes

Liposomes composed of L-α-phosphatidylcholine (PC, chicken egg) (Avanti Polar Lipids, 840051P) lipid and L-α-phosphatidylserine (PS, porcine brain) (Avanti Polar Lipids, 840032 P) were prepared in PBS by applying pulsed ultrasound (Ultrasonic-Homogenizer HTU Soni130, 1 min, pulse on/off: 2 s, amplitude 30%) followed by three 30 min rounds of thermocycling between 4 °C and 56 °C. Liposomes were stored overnight at 4 °C before use. The stock solution had a concentration of 1 mg ml^{–1}.

Isolation of extracellular vesicles from TR146 oral epithelial cells

Extracellular vesicles derived from TR146 oral epithelial cells (hEVs) were isolated via size-exclusion chromatography. The culture medium of at least ten confluent cell culture flasks (75 cm²) was collected after 16 h and centrifuged at 3,000g at 4 °C for 15 min to eliminate dead cells and cell debris. Supernatants were carefully removed and gravity filtered through 0.22 µm PVDF filters (Carl Roth, P666.1). Filtrates were concentrated in pre-washed Amicon Ultra-15 centrifugal filter units with a 100 kDa molecular weight cut-off (Merck, Millipore, UFC9100)

by centrifugation at 3,220g and 4 °C to reach a final volume of 500 µl. Meanwhile, qEV 70 nm size exclusion chromatography columns (Izon, ICO-70) were brought to room temperature and washed with at least 10 ml of filtered PBS (Gibco 10010056). Concentrated samples were added onto the column after the washing step and filled with filtered PBS as the last of the sample entered the column. The first 3 ml of flow-through was regarded as void volume and discarded, followed by collection of a 1.5 ml fraction containing the extracellular vesicles. Columns were washed with at least 10 ml of filtered PBS and stored in 20% ethanol at 4 °C. Isolated extracellular vesicles were kept on ice until further use.

Isolation of *C. albicans*-derived extracellular vesicles

C. albicans-derived (strain SN250, Supplementary Table 1) extracellular vesicles (fEVs) were isolated from biofilm cultures. Biofilms were cultured in a large-scale system using polystyrene roller bottles (Corning, 431133) at 37 °C and 4 r.p.m. for 24 and 48 h⁵². The culture medium was decanted, and remaining debris was removed by filtration through a paper filter. The resulting filtrate was filter sterilized, then concentrated to 25 ml using a Vivaflow 200 unit (Sartorius, VF20H2) equipped with a Hydrosart 30-kDa-cut-off membrane. The concentrate was centrifuged at 10,000g for 1 h at 4 °C to remove larger cellular/membrane-like particulates. The pellets were discarded and the collected supernatants were centrifuged in a Beckman Coulter Optima MAX-XP ultracentrifuge equipped with an MLA-50 fixed rotor at 100,000g for 1.5 h at 4 °C. The resulting supernatant was discarded, and the pellet was resuspended in 5 ml of PBS (pH 7.2) and recentrifuged in the ultracentrifuge utilizing an MLS-50 swinging bucket at 100,000g for 1 h at 4 °C. The collected EVs were then polished by flash size exclusion chromatography on qEV/35 nm columns (Izon Science, SP5102555), filter sterilized and stored at 4 °C⁵³.

Quantification and characterization of EVs by NTA

Purified fEVs were quantified using NTA. Samples were diluted in PBS to a final volume of 1 ml and pretested to obtain an ideal count of 30–100 particles per frame rate using a NanoSight NS300 system equipped with a sample assistant autosampler (Malvern Panalytical). The following settings were applied: camera level was increased to 16 and camera gain to 2 until tested images were optimized and nanoparticles were distinctly visible without exceeding particle signal saturation. Each measurement consisted of five 1 min videos with a delay of 5 s between sample introduction and the start of the first measurement. For detection threshold analysis, the counts were limited to 10–100 red crosses and no more than 5–7 blue crosses. Acquired data were analysed using NanoSight Software NTA v.3.4 Build 3.4.003. At least 1,000 events in total were tracked per sample to minimize data skewing based on single large particles⁵⁴.

Concentration and particle size distribution of hEVs, fEVs and liposome samples were analysed by NTA using a NanoSight NS300 device with a 650 nm laser (Malvern Panalytical). Samples were diluted appropriately in filtered PBS (Gibco 10010056) and kept on ice until measurement. For the measurement, five 1 min videos were captured per sample at 25 °C, a constant sample flow speed of 20, and the camera level and detection threshold set to 14 and 4, respectively.

Liposome and extracellular vesicle treatment with CaL

hEVs (3.57×10^9 particles per ml), PCPS liposomes (1×10^{10} particles per ml) and fEVs (1×10^{11} particles per ml) were incubated with or without 7 µM CaL for 2 h at 37 °C. The size and concentration of the resulting EVs were determined by NTA as described above.

Protection assays

TR146 cells (2×10^4) per well were seeded in 96-well plates in DMEM/F12 (Gibco, 11320074) supplemented with 10% (v/v) heat-inactivated (30 min at 56 °C) fetal bovine serum (Bio&Sell) and cultured until

confluent. PCPS liposomes, hEVs and fEVs were diluted in serum-free DMEM/F12 medium and applied to epithelial cells at a concentration of 1×10^9 particles per well, 2.6×10^8 particles per well and 1×10^9 particles per well, respectively. Synthetic CaL was prepared in serum-free DMEM/F12 and applied at a final concentration of 20 µM. Exhausted culture medium was collected after 4 h and 24 h, and host cell damage was determined using a lactate dehydrogenase (LDH) cytotoxicity detection kit (Roche, 11644793001) according to the manufacturer's instructions.

Characterization of aggregation

Aggregate formation of peptides alone and in combination with CaL was analysed with dynamic light scattering on a Zetasizer Nano ZS90 (Malvern). For measurements of CaL, 1 ml of the 100 µM concentrated peptide in 5 mM HEPES, 100 mM KCl (pH 7) was prepared in a disposable cuvette (4 ml, 67.754, Sarstedt). For measurements of NCEPs and NCEP subfragments (P1, P2, P4, P5, P5a, P5b, P6, P7, P7a, P7b, P8, scrP1, scrP2, scrP4) in combination with CaL, the 1 ml solution comprised 100 µM NCEP and 100 µM CaL. The solution was allowed to equilibrate for 3 min at 37 °C, and the count rate was determined in three technical replicates over 10 cycles of 10 s each. The count rate at high levels of scattering intensity was reduced by an automatic attenuator and the derived attenuation factor was used to determine the count rate as a measure of the extent of aggregate formation. By removal of a volume of the solution and addition of the same volume of buffer, the concentration was gradually decreased to characterize the dissolution of the aggregates. At each concentration step, the count rate was determined as described above. This procedure was repeated at least three times with new peptide solutions, and the mean and standard deviation were determined over the repetitive measurements. For measurements of NCEPs and NCEP subfragments alone (P1, P2, P4, P5, P5a, P5b, P6, P7, P7a, P7b, P8, scrP1, scrP2, scrP4), the concentration was titrated upward by successive additions of 5 µl of peptide solution.

Determination of membrane permeabilization

Tethered bilayer lipid membranes (tBLMs) with 10% tethering lipids and 90% spacer lipids (eDAQ, SDx-T10 slides) were formed using the solvent exchange technique according to manufacturer instructions (SDx Tethered Membranes). Briefly, 8 µl of 3 mM DOPC (Avanti Polar Lipids, 840375 P) solution in ethanol was added, incubated for 2 min and then added with 93.4 µl of buffer (100 mM KCl, 5 mM HEPES, pH 7.0). After rinsing 3 times with 100 µl of buffer, the conductance and capacitance of the membrane were measured for 20 min before injection of Ece1 peptides at different concentrations. All experiments were performed at 37 °C. Signals were measured using TethaPod (SDx Tethered Membranes).

Quantification of membrane permeabilization

Current measurements were performed using multiple planar lipid bilayers and an Orbit e16 system (Nanon) as previously described⁵⁵. Briefly, horizontal bilayers were created using DPhPC lipid (Avanti Polar Lipids, 850356 P) dissolved in octane (25 mg ml^{-1}) together with an electrolyte solution containing 0.1 M KCl and 20 mM HEPES (pH 7.4). CaL dissolved in water was added to the bilayers at final a concentration of 10 µM, in the absence and the presence of P7, P7a and/or P7b at concentrations of 10 µM. A constant voltage of –50 mV was applied, and current changes were monitored for 30 min at room temperature. Current analysis was performed using Clampfit 10.3 (Molecular Devices).

Construction of NCEP deletion and scrambled plasmids for *C. albicans* transformation

Before starting plasmid generation, the signal peptide, conserved KR sites and the correct first amino acid of the following peptide were ensured in the modified *ECE1* sequences to theoretically enable correct

processing. The respective modified *ECE1* sequences were designed and purchased in *E. coli* pUC57 plasmids (Biomatik). The modified *ECE1* sequences were subsequently introduced into the vector plasmid pClp10 + *ECE1* to generate the pClp10-*ECE1*_{ΔscrPx} vectors. To generate pClp10-NCEP deletion constructs, BspI and BsmBI (New England Biolabs, R0585S, R0739S) were used for pUC57-*ECE1*_{ΔPx} plasmid digestion. Dephosphorylation of the vector plasmid was performed to avoid re-ligation using recombinant shrimp alkaline phosphatase. Linearized vector and *ECE1*_{ΔPx} inserts were isolated using a QIAquick gel extraction kit (Qiagen, 28706) according to manufacturer protocol. Ligation of the purified inserts with the linearized pClp10 vector plasmid was performed using T4 DNA ligase. For the scrambled constructs, the pClp10-*ECE1*_{ΔscrPx} plasmids were created without ligation by PCR amplification (Q5 high-fidelity DNA polymerase, New England Biolabs, M0491S) of the pClp10 + *ECE1* vector backbone and the inserts from the respective pUC57-*ECE1*_{scrPx} plasmid (for primers, see Supplementary Table 3; bold complementary sequence between vector backbone and insert). The PCR products were digested with DpnI (New England Biolabs, R0176S) to remove the parental pUC57 plasmid and fragments were directly transformed into *E. coli* (plasmids are formed by the organism)⁵⁶. For *E. coli* transformation, chemically competent DH5α or Stellar competent *E. coli* cells (Takara Bio, 636763) were used. The mixture was incubated on ice (30 min), heat shocked (2 min, 42 °C), SOC medium was added, and the transformation mixture was incubated at 37 °C for 1–1.5 h with shaking. The transformation mixture was plated on selective medium (LB agar + 50 μg ml⁻¹ ampicillin, Roth, HP62.2). Positive clones were re-streaked and plasmids were isolated with a QIAprep Spin miniprep kit (Qiagen, 27106) or GeneJET Plasmid miniprep kit (Thermo Fisher, K0503). Plasmids were verified using PCR, purified using the QIAquick PCR purification kit (Qiagen, 28106) and sequenced before transformation into *C. albicans*.

Transformation of *C. albicans*

Transformation of *C. albicans* was performed as previously described¹⁵ with slight modifications. A uridine auxotrophic *C. albicans* strain lacking *ECE1* (*ece1Δ/Δ* (*ura*⁻), internal number M2042) was transformed using the lithium acetate (LiAc) method⁵⁷. Before transformation, plasmids were linearized with StuI or NcoI HF (New England Biolabs, R0187S and R3193L) and purified by ethanol precipitation. *C. albicans* cells were collected by centrifugation and resuspended in LiAc solution. Denatured salmon sperm carrier DNA (100 μg) (Salmon sperm DNA, sheared (10 mg ml⁻¹), Invitrogen, AM9680), the prepared *C. albicans*-LiAc solution, 10 μg of linearized pClp10 + *ECE1*_{ΔscrPx} plasmid DNA and 600 μl of polyethylene glycol-LiAc solution were combined and mixed to homogeneity. Integration of *ECE1*_{ΔscrPx} constructs into the *C. albicans* *RPS1* locus was confirmed by PCR. Amino acid sequence of Ece1 peptides encoded by the modified *ECE1*_{ΔscrPx} sequences are presented in Supplementary Table 4.

RNA extraction from fungal cells

C. albicans strains were adjusted to 1 × 10⁷ cells per ml in 25 ml RPMI 1640 (hypha-inducing) or 1 × 10⁸ cells per ml (20 min and 3 h) and 1 × 10⁷ cells per ml (24 h) in 8 ml YPD (yeast-maintaining). For hyphal samples, fungal suspensions were distributed in 150 cm² Petri dishes and incubated at 37 °C and 5% CO₂ for 20 min, 3 h or 24 h. Medium and non-adherent cells were removed. *C. albicans* cells were rinsed once with 15 ml ice-cold PBS, loosened with a cell scraper and collected in a tube. For the 24 h timepoint, the medium was not removed due to strong biofilm formation, thus the supernatants were transferred into a collection tube. The residual cells were then treated as described and combined with the respective supernatants. For yeast samples, fungal suspensions were cultured at 30 °C for 20 min, 3 h or 24 h with 180 r.p.m. shaking. Following incubation, the culture was transferred into a tube. The collected cells were centrifuged at 4,000g for 2 min at 4 °C and resuspended in 15 ml fresh, ice-cold PBS. The pellets were

washed with 1 ml ice-cold PBS and centrifuged at 3,000g for 2 min at 4 °C. The supernatant was discarded and pellets were frozen in liquid nitrogen. Fungal RNA was isolated using either the freeze-thawing method⁵⁸ or the RNeasy mini kit (Qiagen, 74106) according to the manufacturer's instructions¹⁶. RNA concentration was determined using a NanoDrop 1000 spectrophotometer (PEQLAB) and quality was confirmed by Bioanalyzer (Agilent) measurements.

cDNA synthesis

RNA (500 ng) was treated with DNase Baseline-ZERO (Lucigen, DB0715K) and cDNA was generated using the Reverse Transcriptase Superscript III system (SuperScript III Reverse Transcriptase 18080085, Oligo(dT)12-18 Primer 18418012, RNaseOUT Recombinant Ribonuclease Inhibitor 10777019) according to respective manufacturer instructions, except that only 0.5 μl reverse transcriptase was used. The cDNA generated was confirmed to be gDNA free by PCR.

Determination and quantification of *HAC1* mRNA splicing for UPR activation

To detect *HAC1* mRNA splicing, quantitative PCR with reverse transcription (RT-qPCR) was performed on cDNA collected from *C. albicans* Wt, Wt treated with 2 μg ml⁻¹ tunicamycin (positive control) and *C. albicans* mutant strains. RT-qPCR was performed with primers specific for *HAC1* (*HAC1*_{SP_F}: CTTCTCCTCATCATCGTTA, *HAC1*_{SP_R}: TCAAAGTCAACTGAAATG) using the following parameters: 95 °C preheat, 95 °C 30 s, 95 °C 30 s, 52 °C 30 s, 68 °C 30 s (33 cycles), 68 °C 5 min. The PCR product (97 bp) was detected using a 3% agarose high resolution gel (ROTH, K297.2) in TBE buffer. Product size was determined using a 25 bp DNA step ladder (Promega, G451A).

To quantify the amount of *HAC1* splicing, gel images were inverted and edited (contrast and brightness only) using Adobe Photoshop (v.24.1.1) with the same settings for all images. Images were subsequently analysed using BioID software (Vilber Lourmat, v.15.08) and the molecular weight analysis tool. Analysis lanes were set to fit the gel lanes and to cover both expected *HAC1* fragments (spliced, *HAC1*^s; unspliced, *HAC1*^u). For each sample, two regions were analysed (located on the same gel), designated *HAC1*^u and *HAC1*^s. Background subtraction was performed using the valley-to-valley analysis and a common threshold was set for all samples in one image. For the final quantification, the area under the curve for both fragments was determined and the percentage of spliced product ((*HAC1*^s/*HAC1*^s + *HAC1*^u) × 100%) was calculated. Values represent the mean of three, and for 1 h and 2 h, four replicates were analysed.

CaL staining

TR146 cells were seeded in 24-well plates (TPP, 92424), grown to confluency and infected with *C. albicans* (1 × 10⁵ cells per well). After 4 h of incubation at 37 °C and 5% CO₂, the samples were washed three times with PBS and fixed with Histofix (Roth, P087.5). Concanavalin A-AlexaFluor 647 (Invitrogen, C21421) was used to stain extracellular, non-invasive fungal components for 20 min at room temperature in the dark on an orbital shaker (70 r.p.m.). Samples were washed twice with PBS and permeabilized with 0.1% saponin in PBS (0.1% PBSS) for 5 min at room temperature and rinsed again twice with PBS. The samples were blocked with 1% BSA in 0.05% PBSS for 1 h at room temperature with 70 r.p.m. shaking, and rinsed once with PBS. All antibodies were diluted in 0.5% BSA in 0.05% PBSS, and all antibody incubations were performed at room temperature for 1 h with shaking at 70 r.p.m., followed by two washing steps with PBS. Anti-CaL VHH CAL1-F1 (1 μM, QVQ, Q100c)¹⁶ was used as the primary antibody, followed by a polyclonal rabbit-α-VHH IgG (1 mg ml⁻¹, QVQ, QE19, 1:50) as a secondary antibody. The primary antibody, defined as a 'single domain antibody' (sdAbs) by QVQ, is termed 'nanobody' in this study. Subsequently, a tertiary antibody was used (goat-α-rabbit IgG coupled with AlexaFluor 488, 2 mg ml⁻¹, Invitrogen, A32731, 1:400)¹⁶. Samples were mounted in

ProLong Gold antifade mounting medium (Invitrogen, P36934) and visualized using fluorescence microscopy. To determine the ratio between CaL-positive and negative hyphae, $\times 20$ magnification was used with Z-stack ($4\ \mu\text{m}$) (Zeiss Observer, ZEN Software v.2.4, ZEISS) and at least 50 hyphae were counted unless the strain exhibited an obvious filamentation defect (*TripleP3n* ≥ 40 and *scr-alln* ≥ 36). The experiment was performed in duplicates as described¹⁶. For the representative staining figures, pictures were taken with $\times 100$ magnification (Zeiss Observer, Zen Software v.2.4, ZEISS) in Z-stack ($4\ \mu\text{m}$) and the x - y projection was cut and adjusted. Channel settings were kept constant between the pictures (ZEN Lite Software v.3.3, ZEISS). Quantification of the CaL staining was performed with $\times 20$ magnification, Z-stack ($4\ \mu\text{m}$) and maximum intensity projection, and the x - y projection (Zeiss Observer, ZEN Software v.2.4, ZEISS) with 10 hyphae per strain and replicate, considering the CaL-positive/negative ratio using the ZEN Software v.3.3 (ZEISS). A fixed area ($8.06\ \mu\text{m} \times 38.70\ \mu\text{m}$) including the hyphal tip was used to determine the mean intensity per hypha and the average thereof was calculated. As the strains *ece1Δ*, $\Delta P3$, $\Delta P3 + 5$, $\Delta P3 + 7$, *scrP3* and *scr-all* do not possess an epitope for the nanobody to bind, the measurements were used to calculate the average background intensity.

Western blot analysis for intracellular Ece1 protein detection

Hyphal samples were prepared as described in 'RNA extraction of fungal cells' and the pellets lysed with 200 μl of extraction buffer containing 0.1 M NaOH, 0.05 M EDTA, 2% SDS and 2% 2-mercaptoethanol (all Roth). Resuspended cells were incubated at $96\ ^\circ\text{C}$ for 10 min and neutralized with 5 μl of neutralization buffer (4 M acetic acid in water)⁵⁹.

SDS-PAGE gels (15%) with a 5% stacking gel were loaded with 45 μl of cell lysate/CaL-loading buffer solution and run at 30 mA. The PageRuler prestained protein ladder (Thermo Scientific, 26620, 10600002) was used as a molecular weight marker. Gels were blotted on nitrocellulose membranes (Amersham Protran0.45 NC, GE Healthcare Life Science) and total protein staining was performed with the Revert 520 Total Protein Stain kit (LI-COR Bioscience, 926-10021). Subsequently, membranes were blocked with 5% milk, and incubated first with 1 μM anti-CaL VHH CAL1-F1¹⁶ in 5% milk, followed by a polyclonal rabbit- α -VHH IgG (1 mg ml^{-1} , QVQ, 1:50) in 5% milk and IRDye 800CW donkey anti-rabbit IgG (LI-COR Bioscience, 926-32213, 1:10,000) in 5% milk. The Odyssey M system (LI-COR Bioscience) was used to visualize the membranes and quantification was performed with Image Studio Lite v.5.2. (LI-COR Bioscience).

Fungal invasion into epithelial cells

C. albicans cells were added to confluent TR146 oral epithelial cells in 24-well plates to a final concentration of 1×10^5 cells per well. Invasion of *C. albicans* into TR146 cells was analysed by differential staining performed according to a previous publication⁵⁸. Briefly, after 4 h of *C. albicans* infection, TR146 cells were washed three times with PBS and fixed with Histofix. Extracellular non-invasive fungal components were stained with concanavalin A-AlexaFluor 647 (Invitrogen, C21421) for 20 min at room temperature in the dark. After washing again three times with PBS, the cells were permeabilized with 0.5% Triton X-100 for 10 min at room temperature and washed again three times with PBS. *C. albicans* cells were stained with calcofluor white (10 $\mu\text{g}\ \text{ml}^{-1}$ in 0.1 M Tris-HCl pH 9.0; Sigma-Aldrich, F3543-1G) for 20 min at room temperature in the dark. Subsequently, samples were washed three times with water, mounted on glass slides with Mowiol-DABCO (Roth, glycerin 3783, Mowiol 4-88 073.1, Tris-HCl 9090, DABCO 0718) and analysed using fluorescence microscopy (Leica DM5500B, Leica Application Suite 2.7.0.9329). The percentage of invasive *C. albicans* hyphae (only calcofluor white stained) was determined from at least 50 counted hyphae per strain from a minimum of three independent experiments.

LC-MS/MS analysis of Ece1 peptides in hyphal culture medium

Analysis of hypha-secreted Ece1 peptides was optimized for the detection of CaL and modified from Moyes et al.¹³. Briefly, *C. albicans* strains were cultured for 18 h under hypha-inducing conditions (yeast nitrogen base medium (Roth, HP26.1) containing 2% sucrose (Roth, 4661), 75 mM MOPS [3-(*N*-morpholino)-2-hydroxypropanesulfonic acid] buffer (Roth, 7101.2) pH 7.2, 5 mM *N*-acetyl-D-glucosamine (Sigma-Aldrich A8625-25G), $37\ ^\circ\text{C}$). Secreted peptides were enriched from the hypha-inducing medium by solid-phase extraction using silica-based C4 (Macherey-Nagel, 730227, Chromabond C4) followed by C18 (Macherey-Nagel 731806, Chromafix) columns, dried in a vacuum concentrator, dissolved in 0.2% formic acid (Carl Roth, 1EHK.1; ROTIPURAN $\geq 99\%$, LC-MS grade) in 71:27.2 (v/v/v) acetonitrile (ACN)-H₂O-dimethyl sulfoxide (DMSO) (Carl Roth, HN40.2; ROTISOLV $\geq 99.98\%$, Ultra LC-MS) and passed through a 10 kDa molecular-weight-cut-off filter (VWR, 516-0229P, 10 kDa MWCO Spin filter with PES membrane) to remove intact proteins. LC-MS/MS analysis was performed using an Ultimate 3000 nano RSLC system (Thermo Fisher) coupled to a QExactive HF mass spectrometer (Thermo Fisher). Peptide separation was performed on an Accucore C4 column (Thermo Fisher PN 16526-157569, $75\ \mu\text{m}\ \text{I.D.} \times 150\ \text{mm}$, $2.6\ \mu\text{m}$) with eluents (A) 0.2% formic acid in 95:5 H₂O-DMSO and (B) 0.2% formic acid in 85:10:5 ACN-H₂O-DMSO using the following gradient: 0–1 min at 50% B and 500 nl min^{-1} , 2 min at 50% B and 400 nl min^{-1} , 2.5 min at 51% B and 200 nl min^{-1} , 6 min at 61% B and 200 nl min^{-1} , 7 min at 64% B and 500 nl min^{-1} , 11–35 min at 75% B and 500 nl min^{-1} , 45–50 min at 96% B and 500 nl min^{-1} , 50.1–65 min at 50% B and 500 nl min^{-1} . The top 15 precursor ions (full scan at m/z 350–1,500, $R = 60\ \text{k}$ (full-width at half-maximum), maximum injection time (IT_{max}) of 200 ms, automatic gain control (AGC) target of 3×10^6 per scan cycle) underwent high-energy collisional dissociation fragmentation at 28% normalized collision energy. Resulting MS2 spectra were monitored at $R = 15\ \text{k}$ (full-width at half-maximum), IT_{max} of 150 ms and an AGC target of 2×10^5 . Proteome Discoverer v.3.0 (Thermo Fisher) and the Sequest HT algorithm were used to search against the protein database of *C. albicans* SC5314 (<https://www.uniprot.org/proteomes/UP000000559>; 03/05/2023 DD/MM/YYYY) including the respective modified Ece1 sequence for NCEP mutant strains. Mass spectra were searched for both non-specific cleavages (no enzyme) and tryptic peptides up to four missed cleavages. Precursor and fragment mass tolerances were 10 ppm and 0.02 Da, respectively. A strict false discovery rate $< 1\%$ (Target Decoy PSM validator node) and a Sequest score (cross correlation, Xcorr) > 3 were required for positive protein/peptide hits. Only rank 1 proteins and peptides of the top scoring proteins were counted. Label-free protein quantification was based on the Minora algorithm of PD3.0 using the precursor abundance based on intensity and a signal-to-noise ratio > 5 .

Ece1 structure prediction

The AlphaFold protein structure database was used to predict the structure of the native Ece1 protein (UniProt Accession number Q07730)^{60,61} and the AlphaFold Colab notebook was used to predict the structures of the modified Ece1 peptides. The *.pdb files obtained were edited with the PyMol2 software (PyMOL Molecular Graphics System, v.2.0) to assign colours to the different peptides and create short video files.

Statistical analysis

All statistical tests were performed using GraphPad Prism v.9.5.1 and v.10.0.2. Values are presented as mean $\pm$ s.d. No statistical methods were used to pre-determine sample sizes, but our sample sizes are similar to those reported in previous publications^{13,16}. Data distribution was assumed to be normal, but this was not formally tested. Statistical tests are indicated in each figure legend. Statistical significance is indicated as follows: * $P \leq 0.05$; ** $P \leq 0.01$; *** $P \leq 0.001$; **** $P \leq 0.0001$; NS, non-significant. Further details can be found in the respective source data files.

Reporting summary

Further information on research design is available in the Nature Portfolio Reporting Summary linked to this article.

Data availability

The data supporting the findings of this study are available within the paper, its supplementary files and source data files. Correspondence and requests for materials should be addressed to the corresponding authors. Source data are provided with this paper.

Code availability

The code to identify single nucleotide polymorphisms used in Fig. 1 is available from Ropars et al.⁴⁶.

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Author contributions

R.M., A.K. and S.A. designed and performed most of the experimental work, analysed most of the data, wrote and edited the manuscript. M.H., T.H., S.M. and D.Y. performed further laboratory work. T.K. performed LC–MS analyses. C.M., M.-E.B. and C.d’E. performed sequence alignments and corresponding data analyses, and designed the corresponding figures. S.G. and T.H. performed peptide aggregation and membrane permeabilization experiments. C.N. and T.G. analysed the peptide aggregation and membrane permeabilization data and wrote the corresponding parts of the manuscript. E.L.P., S.L. and J.P.R. performed Orbit e16 experiments and analysed the corresponding data. V.T. contributed to the experimental design and data analysis of the western blots. R.Z. produced fEVs. C.V. and M.G.B. were involved in the design and execution to the EV protection assays and hEV production. S.M. and L.K. were involved in the experimental design, wrote and edited parts of the manuscript. L.K., J.P.R., M.G.B., C.N., T.G., O.K., T.K., A.A.B. and D.R.A. contributed to editing the manuscript. J.R.N. acquired funding and was involved in the experimental design of the Orbit e16 experiments, writing and editing of the manuscript. B.H. conceived and designed the study, acquired funding, was involved in experimental design, and wrote and edited the manuscript.

Competing interests

The authors declare no competing interests.

Additional information

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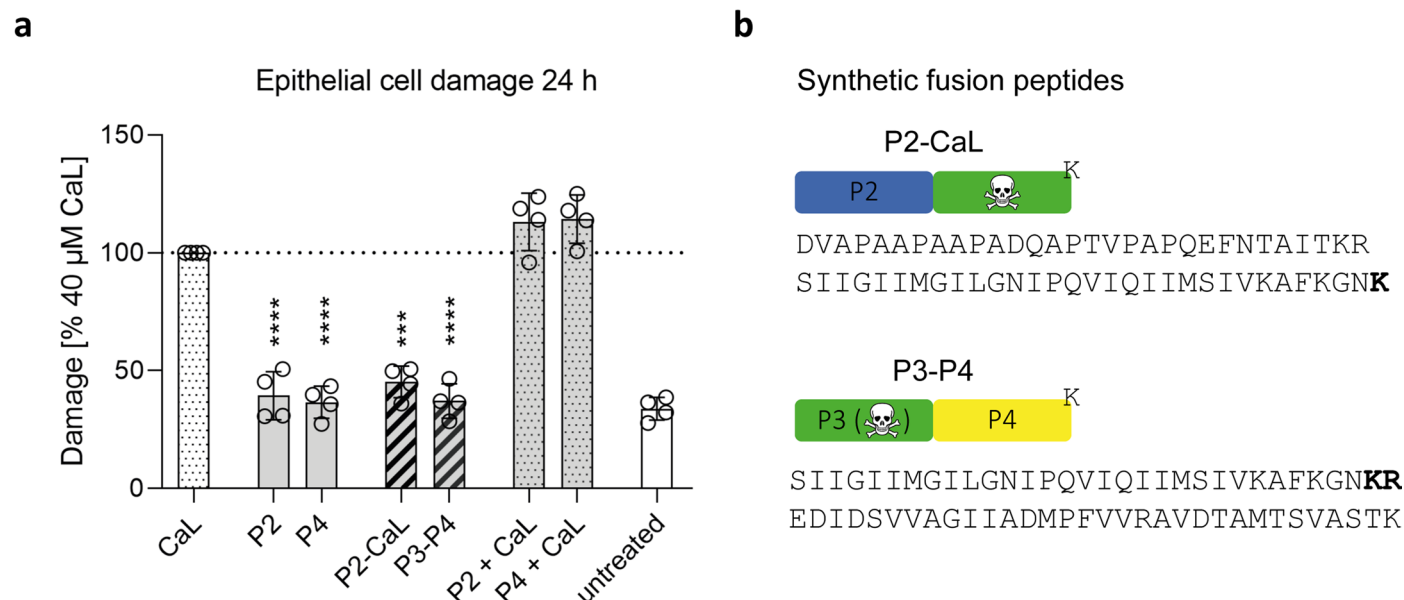
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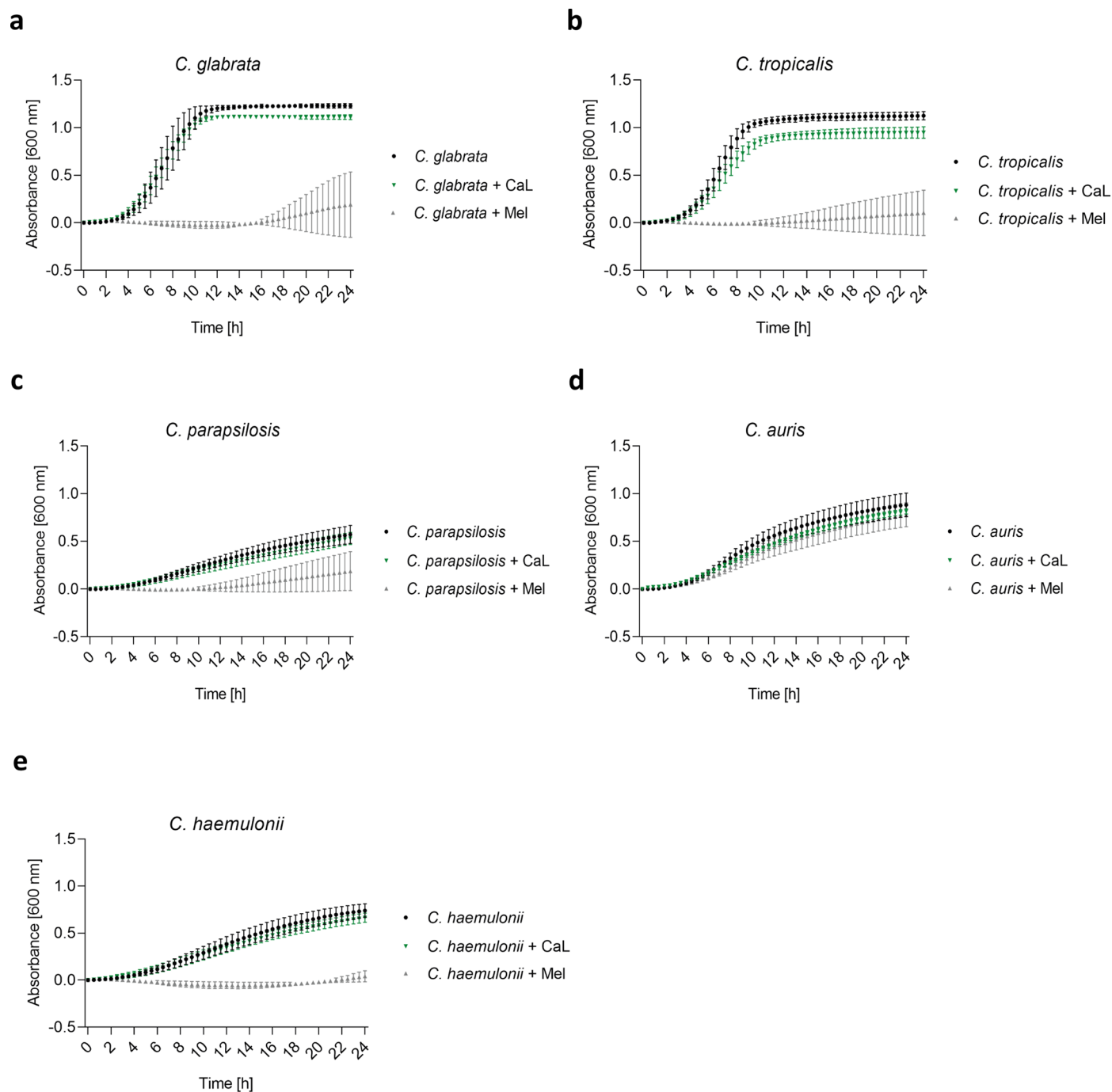
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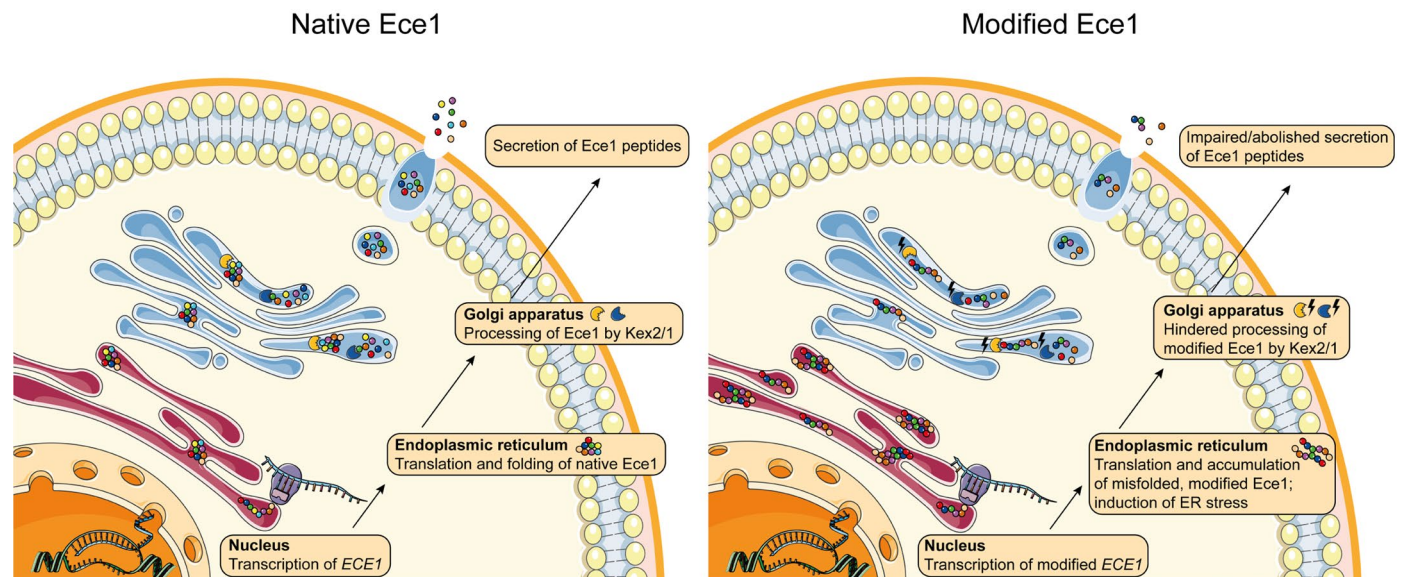
Extended Data Fig. 1 | Synthetic P2-CaL and P3-P4 fusion peptides are non-toxic. a, Damage of oral epithelial cells upon treatment with 40 μM synthetic CaL, synthetic fusion peptides P2-CaL and P3-P4, or a combination of CaL and the respective single peptide for 24 h. Epithelial damage was assessed by quantification of extracellular LDH. Dotted line represents CaL-induced host cell damage (100%). Statistical analysis was performed with one-way ANOVA and Dunnett's multiple comparisons test based on the absorbance data, relative to

the CaL-treated sample, *** $p = 0.001$, **** $p < 0.0001$, $n = 4$ (biological replicates), more statistical parameters are listed in the Source Data Extended Data Fig. 1, mean $\pm$ s.d. **b**, Scheme of the fusion peptides with respective peptide sequence. The fusion peptide P2-CaL ends with K instead of KR, mimicking Kex1 processing of the mature peptide. In fusion peptide P3-P4, the P3 sequence ends with KR to keep the internal Kex2 processing site, whereas P4 ends with K to mimic the Kex1-processed mature peptide.



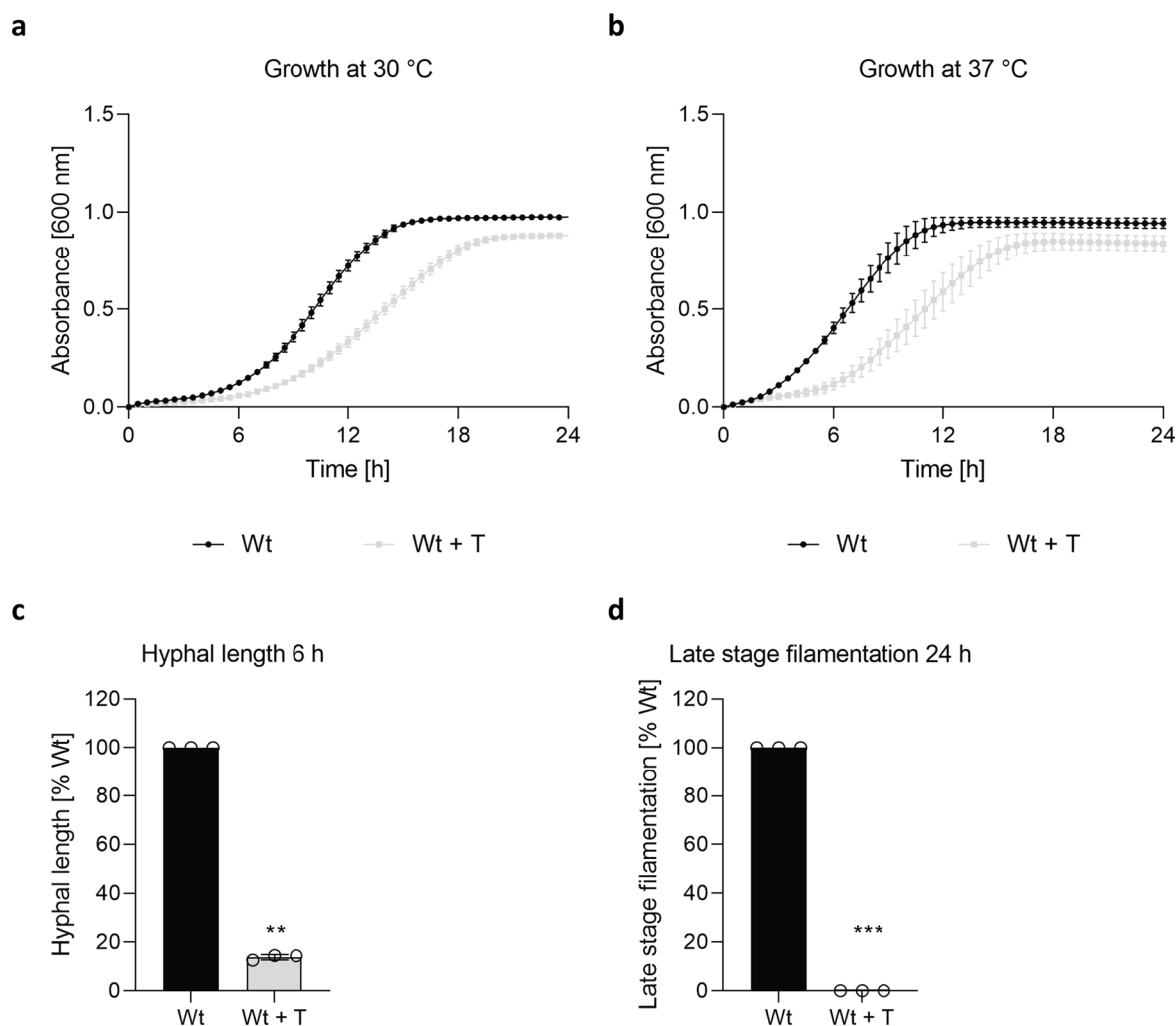
Extended Data Fig. 2 | Susceptibility of different *Candida* species to candidalysin. a–e, Growth of **a**, *Candida glabrata* (ATCC2001) **b**, *Candida tropicalis* (DSM4959) **c**, *Candida parapsilosis* (GA1)⁵⁰ **d**, *Candida auris* (B8441)⁵¹ **e**, *Candida haemulonii* (CBS 5149 T) with or without addition of 70 μ M synthetic

candidalysin (CaL) or synthetic melittin (Mel). Yeast growth was monitored for 24 h by measuring the absorbance (600 nm) every 30 min. a–e n = 3 (biological replicates), mean $\pm$ s.d.



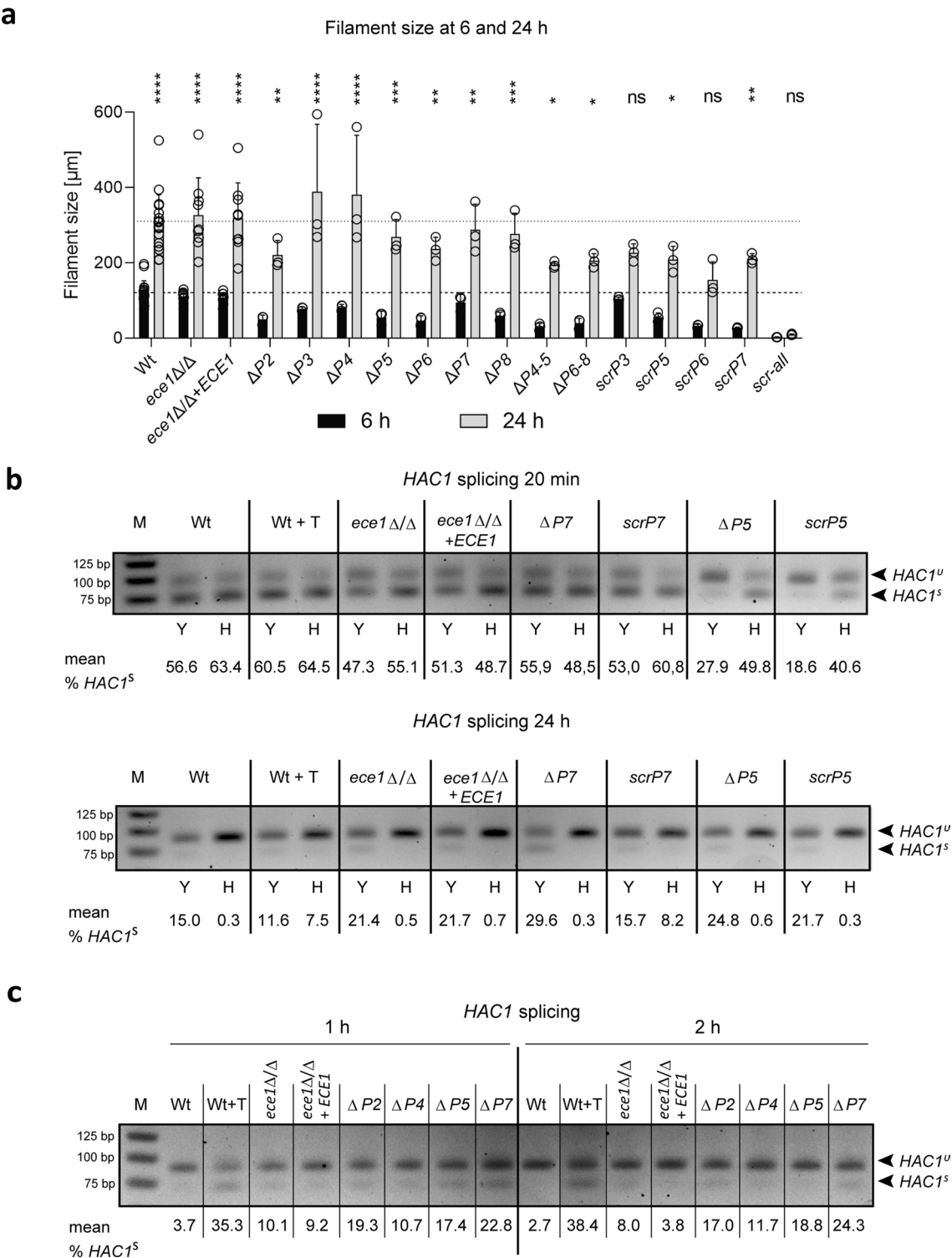
Extended Data Fig. 3 | Schematic of how Ece1 modification (NCEP modifications or deletions) promotes ER stress and an unfolded protein response. Left: In Wt *C. albicans* cells, transcription of native *ECE1* is followed by translation and folding of Ece1 in the ER, transport to the Golgi apparatus and sequential processing by Kex2 and Kex1 proteases, enabling candidalysin and

NCEP secretion. Right: In NCEP mutants, the modified Ece1 structure promotes misfolding which triggers ER stress. Ece1 modifications hinder correct Kex2/1 processing leading to abolished or impaired secretion of candidalysin. The figures were drawn using pictures from Servier Medical Art by Servier, licensed under a Creative Commons Attribution 3.0 Unported License.



Extended Data Fig. 4 | The ER stressor tunicamycin inhibits hyphal growth. **a+b**, Growth of a *C. albicans* Wt strain at 30 °C (**a**) and 37 °C (**b**) in YPD supplemented with tunicamycin (2 µg/mL (Wt + T)). Growth was monitored for 24 h by measuring the absorbance (600 nm) every 30 min. Data were normalized following subtraction of the first measurement (baseline), biological replicates $n = 3$, mean $\pm$ s.d. **c+d**, Hyphal length of Wt *C. albicans* with tunicamycin treatment (2 µg/mL) after 6 h (**c**, mean Wt hyphal length: $125.6 \mu\text{m} \pm 9.4 \mu\text{m}$) or microcolony

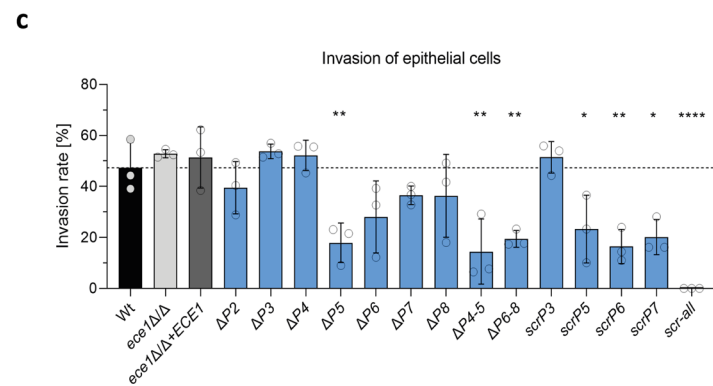
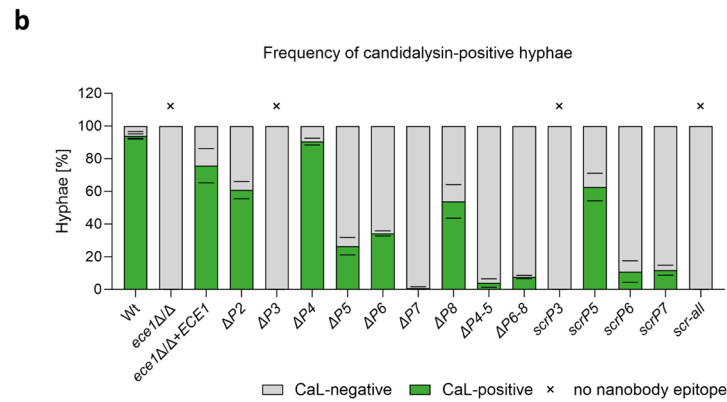
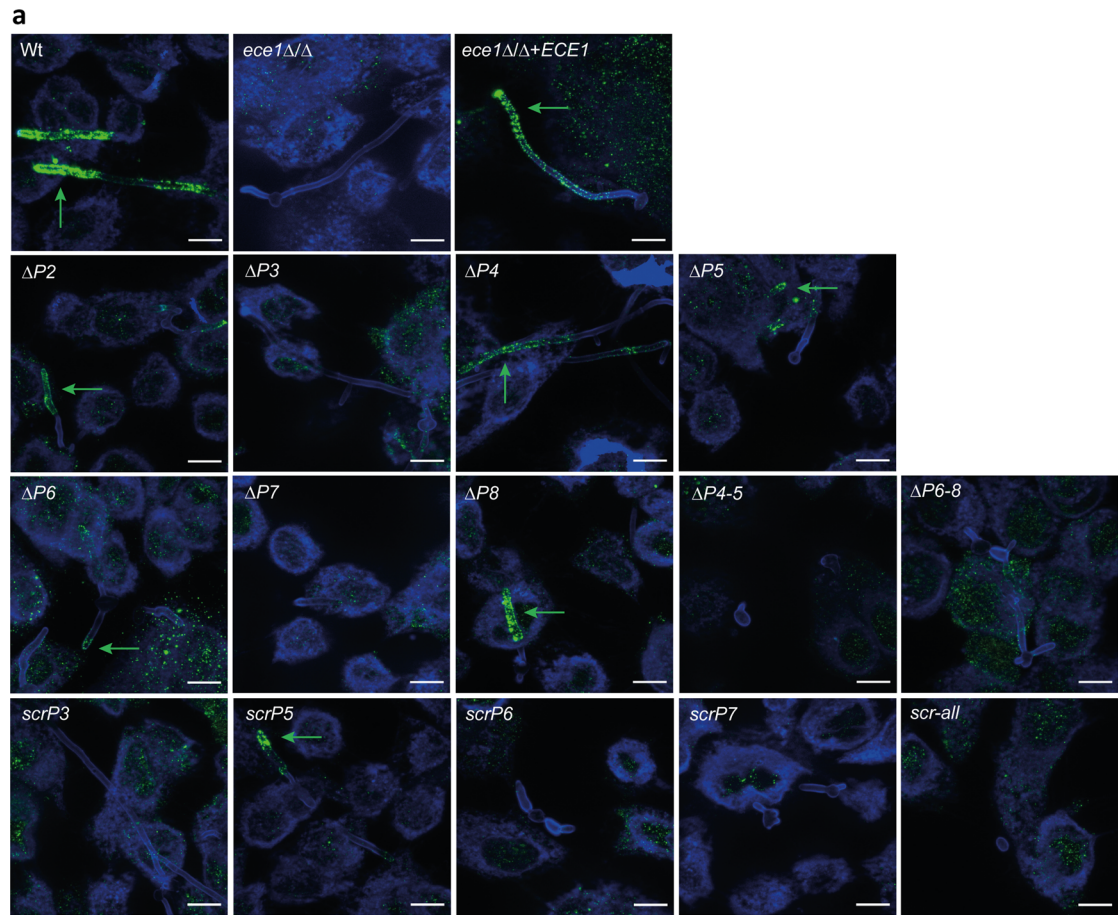
diameter of the Wt with tunicamycin treatment (2 µg/mL) after 24 h (**d**, mean Wt microcolony diameter: $404.4 \mu\text{m} \pm 21.10 \mu\text{m}$). Hyphal length and microcolony diameter were determined microscopically by measuring at least 50 hyphae or 20 microcolonies per replicate. Statistical analysis was performed with a paired t-test, two-tailed, on the raw µm values, c, ** $p = 0.0015$ and d, *** $p = 0.0009$ $n = 3$, mean $\pm$ s.d.



Extended Data Fig. 5 | See next page for caption.

Extended Data Fig. 5 | Altered hyphal formation, UPR dynamics, and UPR quantification in response to Ece1 modifications. Ece1 sequence variations lead to reduced hypha formation but do not affect hypha formation *per se*, and ER stress induced by Ece1 modification is ameliorated over time. **a**, Filamentation of Wt (n = 18), *ece1Δ/Δ* (n = 9), *ece1Δ/Δ+ECE1* (n = 9) complemented strain, a mutant strain lacking the candidalysin-encoding sequence only ($\Delta P3$), NCEP deletion mutants, and scrambled peptide mutants after 6 h (hyphal length, black, n = 3) or 24 h (microcolony diameter, grey, n = 3). Black dotted line represents Wt-level after 6 h of hypha induction, grey dotted line represents Wt-level after 24 h of hypha induction, n represents biological replicates. Hyphal length and late stage filamentation were determined microscopically measuring at least 50 hyphae or 20 microcolonies per replicate (*scr-all* 24 h ≥ 12). Analysis was performed with two-way ANOVA and a Šidák's multiple comparisons test of every 6 h value relative to the respective 24 h value. * $p \leq 0.05$, ** $p \leq 0.01$, *** $p \leq 0.001$, **** $p < 0.0001$, ns = non-significant, more statistical parameters are listed in the Source Data Extended Data Fig. 5, mean $\pm$ s.d. Values for the 6 h time point are also presented in Fig. 4b as % of the Wt and shown here as absolute values

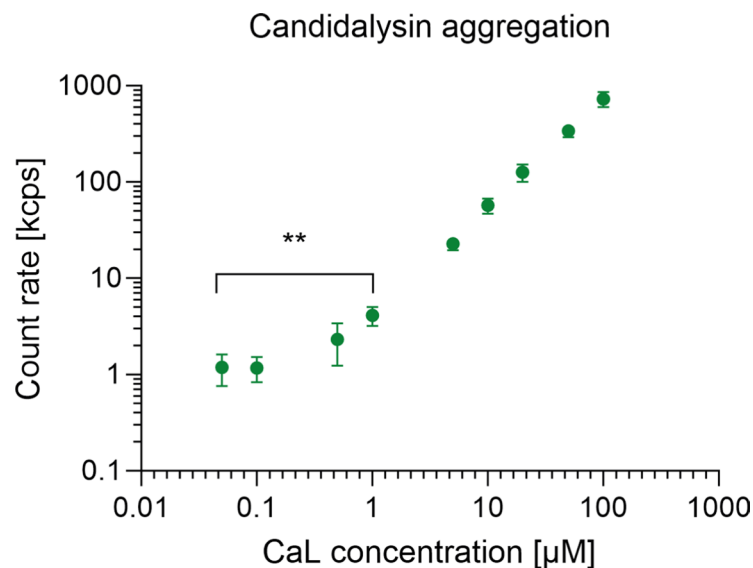
for comparison against the size of late stage filamentation size. **b+c**, Analysis and quantification of *HACI* mRNA splicing as a read-out for ER stress and UPR induction. Activation of a UPR was determined by visualization of *HACI* mRNA splicing by RT-qPCR after culturing the indicated *C. albicans* strains for **b**, 20 min and 24 h, and **c**, 1 h and 2 h under yeast- and/or hypha-inducing conditions. A band at 97 bp shows *HACI* mRNA splicing, indicating ER stress and UPR activation Y = yeast growth conditions (30 °C, 180 rpm shaking); H = hyphal growth conditions (37 °C, 5 % CO₂, plastic surface); positive control: Wt treated with tunicamycin (2 μ g/mL (Wt + T)). One representative replicate from 3 biological repeats per time point is shown. Quantification of *HACI* mRNA splicing is presented as the percentage of spliced *HACI* mRNA from total *HACI* mRNA level quantified with Bio1D from Vilber Lourmat. **b**, n = 3 and **c**, n = 4. A graphical representation of quantified *HACI* mRNA splicing for all replicates including standard deviation and statistics is presented in Supplementary Fig. 2. In **c**, 2 h Wt+T and Wt bands derived from the same gel were cropped and swapped in position to maintain a consistent sample order. Cropped regions are indicated in the Extended_Data_Figure_5_1h_2h_all_replicates_source_data_gels file, with a frame.



Extended Data Fig. 6 | See next page for caption.

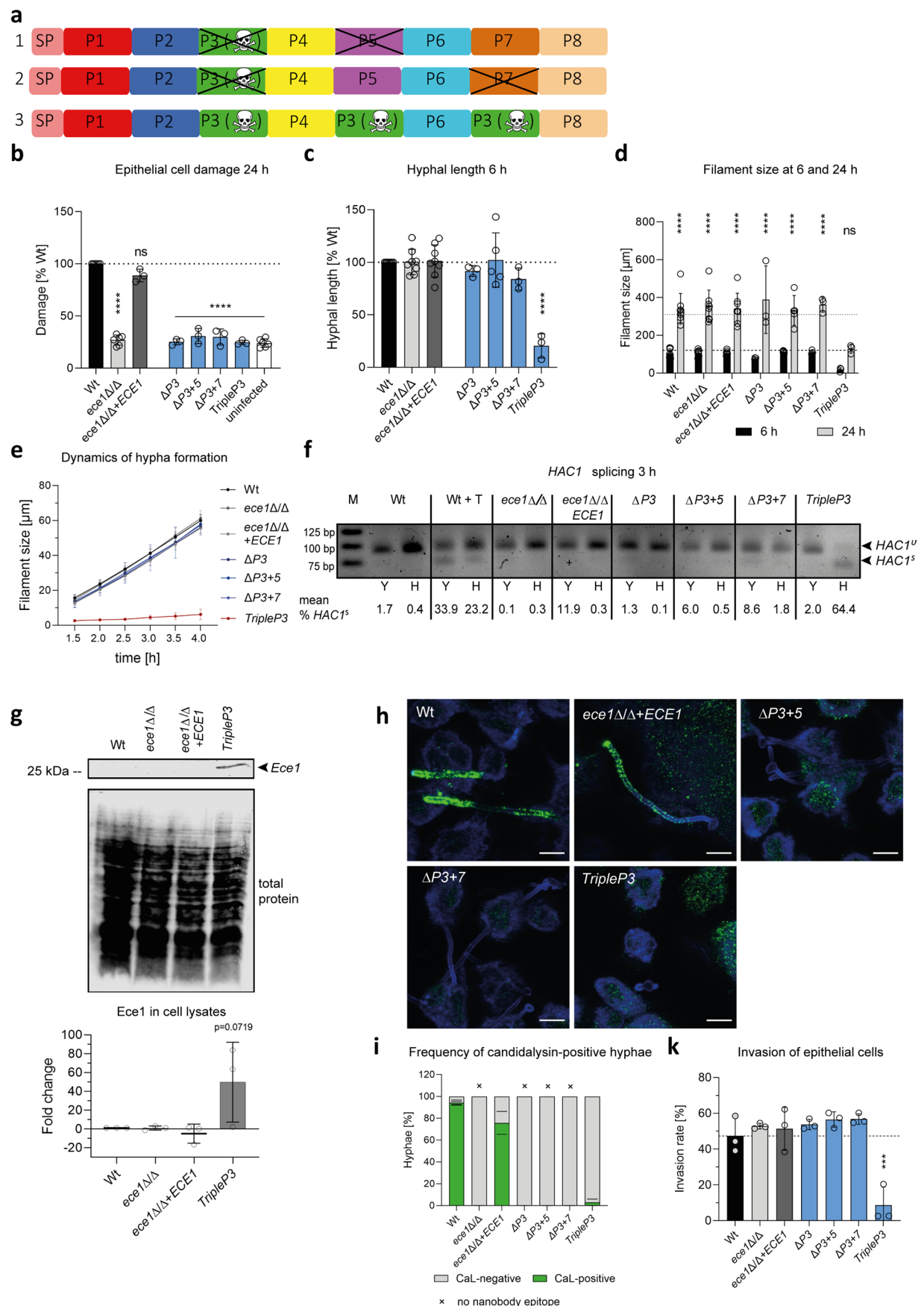
Extended Data Fig. 6 | Candidalysin secretion is impaired in NCEP mutants. **a**, Representative pictures of candidalysin (CaL) staining for each tested strain. CaL was stained by anti-CaL nanobody VHH CAL-F1 (green)³ and extracellular fungal regions by Concanavalin A-AlexaFluor 647 (blue) after infecting TR146 oral epithelial cells for 4 h. Green arrows indicate CaL staining, white scale bars represent 10 μ m. Quantification of the mean fluorescence intensity of each strain is presented in Supplementary Fig. 3. *ece1* Δ/Δ , Δ *P3*, *scrP3*, and *scr-all* lack a native candidalysin epitope. **b**, The numbers of CaL-positive (green) or CaL-negative (grey) hyphae were determined and the percentage of CaL-positive or CaL-negative hypha was calculated. At least 50 hyphae from two independent

replicates were analyzed per mutant (except the *scr-all* mutant (hyphae ≥ 36) which was predominantly yeast cells, and the *scr7* mutant (hyphae ≥ 47)), $n = 2$ biological replicates, bar presents mean. **c**, Invasion of NCEP mutants into oral epithelial cells presented as invasion rate (%). Invasion was determined microscopically after differential staining of intra- and extracellular fungal parts for at least 50 hyphae per strain and replicate ($n = 3$). Statistical analysis was performed with one-way ANOVA and a Dunnett's multiple comparisons test relative to the Wt sample, mean $\pm$ s.d. * $p \leq 0.05$, ** $p \leq 0.01$, **** $p < 0.0001$, more statistical parameters are listed in the Source Data Extended Data Fig. 6.



Extended Data Fig. 7 | Synthetic candidalysin aggregates in aqueous solution. Candidalysin aggregation was determined in aqueous solution as the count rate of aggregates in a concentration range between 0.05 μM and 100 μM by DLS. Statistical analysis was performed with an unpaired, two-tailed t-test

relative to the lowest CaL concentration (0.05 μM) with a statistically significant change starting at 1 μM . $n = 3$ experiments, ** $p = 0.0072$, mean $\pm$ s.d., more statistical parameters are listed in the Source Data Extended Data Fig. 7.



Extended Data Fig. 8 | See next page for caption.

Extended Data Fig. 8 | Candidalysin is the driver of ER stress and UPR in NCEP mutants.

a, Schematic view of NCEP mutants produced for this experiment: 1. Deletion of P3 in combination with P5; 2. Deletion of P3 in combination with P7; 3. Three copies of P3 replacing P5 and P7; **b**, Damage of TR146 oral epithelial cells 24 h post infection with Wt ($n = 6$), *ece1Δ/Δ* ($n = 6$), *ece1Δ/Δ+ECE1* complemented strain, a mutant strain lacking the candidalysin-encoding sequence only ($\Delta P3$), mutants lacking the Cal-encoding sequence in combination with P5 or P7 ($\Delta P3 + 5$, $\Delta P3 + 7$), and a mutant harbouring three copies of Cal (*TripleP3*) ($n = 3$). Data are presented as % damage relative to the Wt. Host cell damage was assessed by quantification of extracellular LDH. Statistical analysis was performed with a Mixed-effects analysis and a Dunnett's multiple comparisons test, two tailed, relative to the Wt sample on raw absorbance values, *** $p \leq 0.001$, **** $p \leq 0.0001$, n represent biological replicates, uninfected $n = 6$. **c**, Hyphal length of the indicated *C. albicans* strains after 6 h of hypha induction. Data are presented as % Wt hyphal length (mean Wt = $104.5 \mu\text{m} \pm 23.9 \mu\text{m}$). Hyphal length was determined microscopically measuring at least 50 hyphae per replicate. Statistical analysis was performed with one-way ANOVA and a Dunnett's multiple comparisons test relative to the hyphal length of the Wt sample (μm). **** $p \leq 0.0001$, Wt, *ece1Δ/Δ* and *ece1Δ/Δ+ECE1* $n = 8$, $\Delta P3$, $\Delta P3 + 7$ and *TripleP3* $n = 3$, $\Delta P3 + 5$ $n = 5$, n represents biological replicates. **d**, Filamentation of indicated *C. albicans* strains after 6 h (hyphal length, black) or 24 h (microcolony diameter, grey). Black dotted line represents Wt-level after 6 h ($104 \mu\text{m} \pm 23.9 \mu\text{m}$) of hypha induction, grey dotted line represents Wt-level after 24 h ($340.6 \mu\text{m} \pm 80.7 \mu\text{m}$) of hypha induction. Hyphal length and late stage filamentation were determined microscopically measuring at least 50 hyphae or 20 microcolonies per strain and replicate. Analysis was performed with two-way ANOVA and a Šidák's multiple comparisons test of every 6 h value relative to the respective 24 h value. **** $p \leq 0.0001$, ns = non-significant, Wt, *ece1Δ/Δ* and *ece1Δ/Δ+ECE1* $n = 8$, $\Delta P3$, $\Delta P3 + 7$ and *TripleP3* $n = 3$, $\Delta P3 + 5$ $n = 5$, n represents biological replicates. Some of the values for the 6 h time point are also presented in Fig. 4b as % of the Wt and are shown here as absolute values for comparison against the size of late stage filamentation. **e**, Dynamics of hypha formation of indicated *C. albicans* strains was measured from 1.5 h to 4 h every 30 min on plastic at 37 °C 5 % CO₂ ($n = 3$ biological replicates, ≥ 20 hyphae/replicate were tracked over time, except *TripleP3*: $n = 2$ for 1.5 h and 2 h, therefore no s.d.). **f**, Analysis of *HAC1* mRNA splicing as a read-out for ER stress and UPR induction. Activation of the UPR was determined by visualization of *HAC1* mRNA splicing by RT-qPCR after culturing the respective *C. albicans*

strains for 3 h under yeast- or hypha-inducing conditions. A band at 97 bp shows *HAC1* splicing, indicating ER stress and UPR activation. Quantification of *HAC1* mRNA splicing is given as percentage (mean out of 3 biological replicates, quantified with BioID from Vilber Lourmat, mean values, standard deviations and statistics are provided in Supplementary Fig. 2). Y = yeast growth conditions (30 °C, 180 rpm shaking); H = hyphal growth conditions (37 °C, 5 % CO₂, plastic surface); positive control: Wt treated with tunicamycin (2 $\mu\text{g}/\text{mL}$ (Wt + T)). One representative replicate from 3 biological repeats is shown. **g**, Western blot analysis of hyphal lysates (3 h hyphal induction, plastic, 37 °C and 5 % CO₂) indicates the accumulation of *TripleP3* Ece1 polypeptide. Quantification was performed using Image Studio Lite Version 5.2. and normalization was performed based on total protein staining ($n = 3$ biological replicates). Statistical analysis was performed with one-way ANOVA and a Dunnett's multiple comparisons test relative to the Wt sample. **h**, Representative pictures of candidalysin staining are shown for each mutant strain and in comparison, for the Wt and *ece1Δ/Δ+ECE1* complemented strain. Candidalysin was stained by anti-candidalysin nanobody VHH CAL-F1 (green)³⁶ and extracellular fungal regions by Concanavalin A-AlexaFluor 647 (blue) after infecting TR146 oral epithelial cells for 4 h. White scale bars represent 10 μm . As expected, Cal antigens were not detected in mutants lacking native Cal sequences ($\Delta P3 + 5$, $\Delta P3 + 7$). Quantification of the mean fluorescence intensity of each strain is presented in Supplementary Fig. 3. **i**, The number of Cal-positive (green) or Cal-negative (grey) hyphae was determined and the percentage of Cal-positive or Cal-negative hypha was calculated. At least 50 hyphae were analyzed per mutant (except *TripleP3*: $n \geq 40$, predominantly yeast cells) and replicate after fluorescent staining, $n = 2$ biological replicates. **k**, Invasion of NCEP mutants into oral epithelial cells given as invasion rate in percentage. Invasion was determined microscopically after differential staining of intra- and extracellular fungal parts for at least 50 hyphae per strain and replicate. Statistical analysis was performed with one-way ANOVA and a Dunnett's multiple comparisons test relative to the Wt sample ($n = 3$, *** $p \leq 0.001$). **b-k**, Data for Wt, *ece1Δ/Δ*, *ece1Δ/Δ+ECE1* complemented strain, and the mutant strain lacking the candidalysin-encoding sequence only ($\Delta P3$) have already been displayed in Fig. 3, Fig. 4b+c, Extended Data Fig. 5a, and Extended Data Fig. 6 and are shown here for comparison. All panels, except i, mean $\pm$ s.d., and more statistical parameters can be found in the Source Data Extended Data Fig. 8.

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For all statistical analyses, confirm that the following items are present in the figure legend, table legend, main text, or Methods section.

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- ☐ ☒ The exact sample size (n) for each experimental group/condition, given as a discrete number and unit of measurement
- ☐ ☒ A statement on whether measurements were taken from distinct samples or whether the same sample was measured repeatedly
- ☐ ☒ The statistical test(s) used AND whether they are one- or two-sided
Only common tests should be described solely by name; describe more complex techniques in the Methods section.
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- ☐ ☒ A description of any assumptions or corrections, such as tests of normality and adjustment for multiple comparisons
- ☐ ☒ A full description of the statistical parameters including central tendency (e.g. means) or other basic estimates (e.g. regression coefficient) AND variation (e.g. standard deviation) or associated estimates of uncertainty (e.g. confidence intervals)
- ☐ ☒ For null hypothesis testing, the test statistic (e.g. F , t , r) with confidence intervals, effect sizes, degrees of freedom and P value noted
Give P values as exact values whenever suitable.
- ☒ ☐ For Bayesian analysis, information on the choice of priors and Markov chain Monte Carlo settings
- ☒ ☐ For hierarchical and complex designs, identification of the appropriate level for tests and full reporting of outcomes
- ☒ ☐ Estimates of effect sizes (e.g. Cohen's d , Pearson's r), indicating how they were calculated

Our web collection on [statistics for biologists](#) contains articles on many of the points above.

Software and code

Policy information about [availability of computer code](#)

Data collection

Growth curve and absorbance data was collected using Tecan i-control v 2.0.10.0. Microscopic images were acquired with an Observer or Axio Vert microscope using Zen Software 2.4 and with a Leica DM5500B microscope and Leica Application Suite 2.7.0.9329. NTA measurements were conducted with a NanoSight NS300 device. Conductance and capacitance of DOPC membranes were measured using the TethaPod (SDx Tethered Membranes Pty Ltd). Current measurements of DPhPC membranes were performed using an Orbit e16 system (Nanon). LC-MS/MS analysis was carried out on an Ultimate 3,000 nano RSLC system (Thermo Fisher Scientific) coupled to a QExactive HF mass spectrometer (Thermo Fisher Scientific). Odyssey M system and LI-COR Acquisition Software (LI-COR Bioscience) was used to visualize the western blot membranes.

Data analysis

Data were analysed using Graphpad prism v 9.5.1. and v 10.0.2. Microscopic images analysis was performed using ZEN lite Software 3.3 blue edition. The AlphaFold protein structure database was used to predict the structure of the native Ece1 protein (UniProt Accession number Q07730) and the AlphaFold Colab notebook was used to predict the structures of the modified Ece1 peptides. Obtained *.pdb files were edited with the PyMol2 software (The PyMOL Molecular Graphics System, Version 2.0 Schrödinger, LLC.). ECE1 gene analyzes was performed with PHASE v2.1.1. NTA measurements were analyzed with NanoSight Software NTA 3.4 Build 3.4.003. Aggregate formation of peptides was analyzed using a zetasizer Nano ZS90. Current analysis was performed using Clampfit 10.3 (Molecular Devices). Western blot quantification was performed with the Image Studio Lite Version 5.2. (LI-COR Bioscience). Gel pictures were processed and analyses with Adobe Photoshop (Version 24.1.1) and Bio1D software (Vilber Lourmat, Version 15.08) and the molecular weight analysis tool. Proteome Discoverer 3.0 (Thermo Fisher Scientific) and the Sequest HT algorithm were used to search against the protein database of *C. albicans* SC5314 (<https://www.uniprot.org/proteomes/UP000000559>; 2023/05/03 YYYY/MM/DD). Label-free protein quantification was based on the Minora algorithm of Proteome Discoverer 3.0 using the precursor abundance based on intensity and a signal-to-noise ratio >5. Aggregate formation of peptides were analyzed with dynamic light scattering on a zetasizer Nano ZS90 (Malvern)

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Data

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All manuscripts must include a [data availability statement](#). This statement should provide the following information, where applicable:

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The authors declare that the data supporting the findings of this study are available within the paper and its supplementary information files/ source data files. The protein database of *C. albicans* SC5314 (<https://www.uniprot.org/proteomes/UP000000559>; 2023/05/03 YYYY/MM/DD) was used for LC-MS/MS analysis. *C. albicans* genome sequence available from the Candida genome database (version A22-s07-m01-r130) was used. All relevant data are available upon request from the authors.

Human research participants

Policy information about [studies involving human research participants and Sex and Gender in Research](#).

Reporting on sex and gender	The study is exclusively based on in vitro experiments (without human samples).
Population characteristics	Not relevant for the study
Recruitment	Not relevant for the study
Ethics oversight	Not relevant for the study

Note that full information on the approval of the study protocol must also be provided in the manuscript.

Field-specific reporting

Please select the one below that is the best fit for your research. If you are not sure, read the appropriate sections before making your selection.

☒ Life sciences ☐ Behavioural & social sciences ☐ Ecological, evolutionary & environmental sciences

For a reference copy of the document with all sections, see [nature.com/documents/nr-reporting-summary-flat.pdf](https://www.nature.com/documents/nr-reporting-summary-flat.pdf)

Life sciences study design

All studies must disclose on these points even when the disclosure is negative.

Sample size	Sample size of each experiment is stated in Materials and Methods and Figure legends. No sample size calculations were performed and sample sizes were chosen comparable to previously performed studies. E.g. Mogavero et al. 2021, Moyes et. al 2016.
Data exclusions	No data were excluded, unless quality control controls warranted exclusion.
Replication	All experiments were independently repeated at least three times, unless otherwise stated. Details for each experiment are provided in the respective figure legends and in the Materials and Methods section.
Randomization	Randomization was not relevant to this study, as all experiments were performed in vitro. Fungi and host cells were cultured under identical conditions and unbiasedly allocated to well positions and treatments. Samples were harvested, processed and analyzed in random order wherever possible. Microscopic hyphal length measurements was performed by randomly chosen hyphae for quantification. Locations for

representative microscopy images were selected in a random place using Zeiss Zen software 3.3 blue edition, and processed in the same settings.

Blinding All data obtained were from in vitro experiments. Some experiments (e.g. sequence alignments, growth curves) excluded the option of any bias. Quantification (e.g. damage data, HAC1 mRNA splicing) were performed without subjective pre-selection and based on automated analysis. Quantifications of subsamples (e.g. hyphal length) were based on random selections. No subjective quantification has been performed.

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Materials & experimental systems

n/a Involved in the study

☐ ☒ Antibodies

☐ ☒ Eukaryotic cell lines

☒ ☐ Palaeontology and archaeology

☒ ☐ Animals and other organisms

☒ ☐ Clinical data

☒ ☐ Dual use research of concern

Methods

n/a Involved in the study

☒ ☐ ChIP-seq

☒ ☐ Flow cytometry

☒ ☐ MRI-based neuroimaging

Antibodies

Antibodies used

anti-candidalysin VHH CAL1-F1 (QVQ, Cat. Nr. Q100c),
polyclonal rabbit-anti-VHH IgG (QVQ, Cat. Nr. QE19),
goat- α -rabbit IgG coupled with AlexaFluor 488 (Invitrogen, A32731)
IRDye® 800CW Donkey anti-Rabbit IgG Secondary Antibody (LI-COR Bioscience Selected P/N: 926-32213)

Validation

Primary antibody (nanobody): anti-candidalysin VHH CAL1-F1, described in: Mogavero, S. et al. Candidalysin delivery to the invasion pocket is critical for host epithelial damage induced by *Candida albicans*. Cell Microbiol 23, e13378, doi:10.1111/cmi.13378 (2021).
Secondary antibodies: Validation performed by the manufacturer (QVQ, ThermoFisher Scientific, LI-COR Bioscience)

Eukaryotic cell lines

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Cell line source(s)

Human oral epithelial cells (TR146) were used throughout this study. The cell line was purchased from the European Collection of Authenticated Cell Cultures.

Authentication

Cell lines have been authenticated via commercial STR profiling (Eurofins Genomics Europe Applied Genomics GmbH).

Mycoplasma contamination

Mycoplasma contamination was routinely checked previously and the results were negative.

Commonly misidentified lines
(See [ICLAC](#) register)

No misidentified cell lines used.